

Statistics Essentials for Data Science



Inferential Statistics



Learning Objectives

By the end of this lesson, you will be able to:

- 👁 Explain the p-value
- 👁 Examine the different types of hypothesis testing outcomes
- 👁 Analyze concepts such as confidence intervals and significance levels
- 👁 Explain one-tail and two-tail tests



Business Scenario

ABC is an organization in the healthcare industry. The organization has developed an application to manage the data of all its hospitals, medical colleges, and institutions.

However, when the organization installed the application, it faced issues such as the application hanging or crashing while in use. The organization wants to perform testing on the application. It has planned to do hypothesis testing and evaluate other factors, such as peak hours, overloaded servers, and hardware incompatibilities, that will help enhance the performance of the application.

To do so, it should explore hypothesis testing, significance levels, and one-tail tests and determine their usefulness.



Discussion: Inferential Statistics

Duration: 15 minutes

- What is inferential statistics?
- What are the different types of inferential statistics?





Inferential Statistics

Inferential Statistics

It refers to a branch of statistics that deals with making inferences or predictions about a population based on sample data.

It involves utilizing statistical tools to draw conclusions about an entire group based on data collected from a part of that group.

It is employed when studying an entire population becomes impractical or impossible due to time and resource constraints.

Inferential Statistics

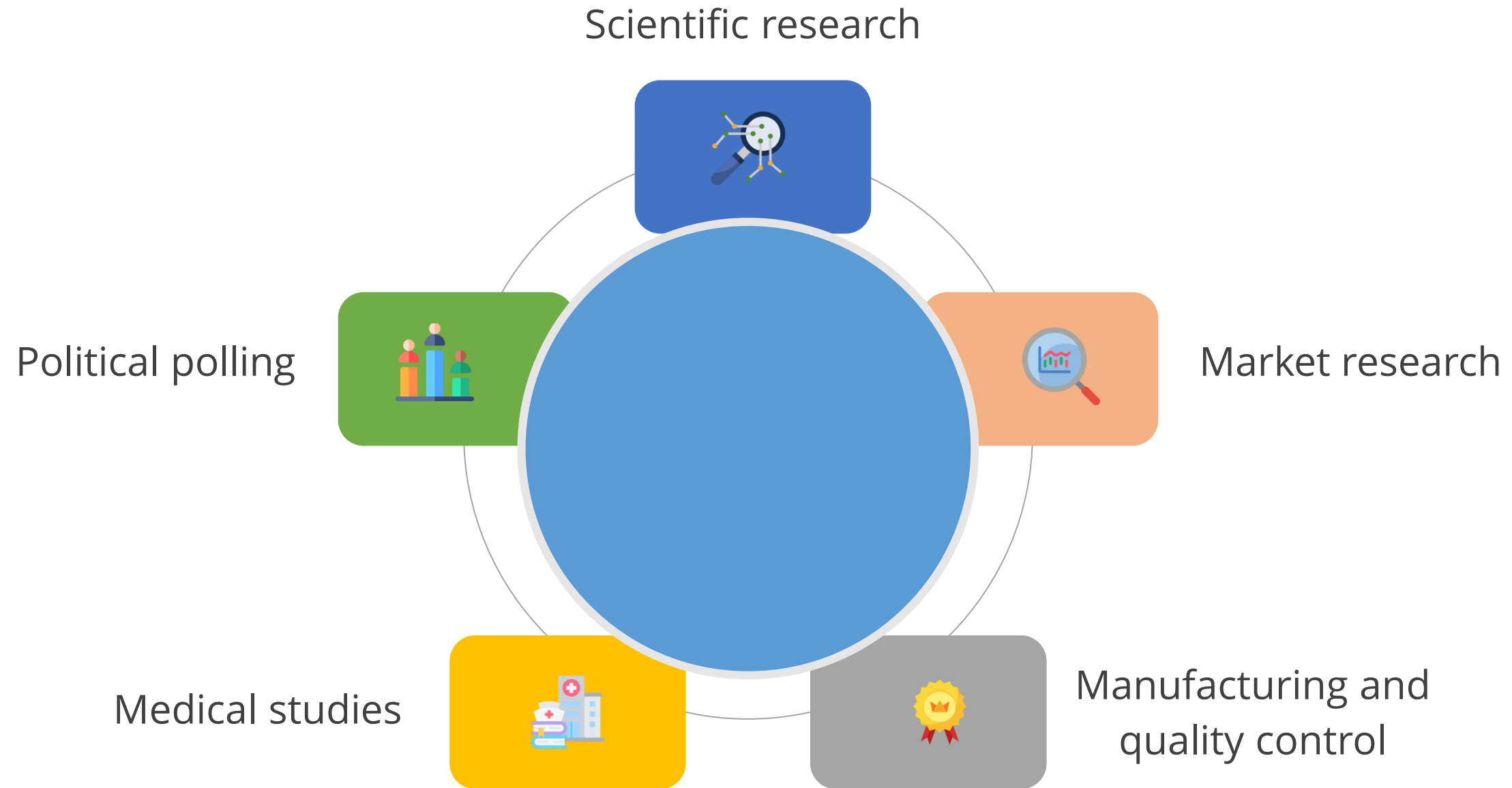
Consider the task of determining the average income of residents in a city



- It is not feasible to collect data from every resident.
- In such cases, collect data from a sample of residents and use inferential statistics to estimate income levels for the entire population.

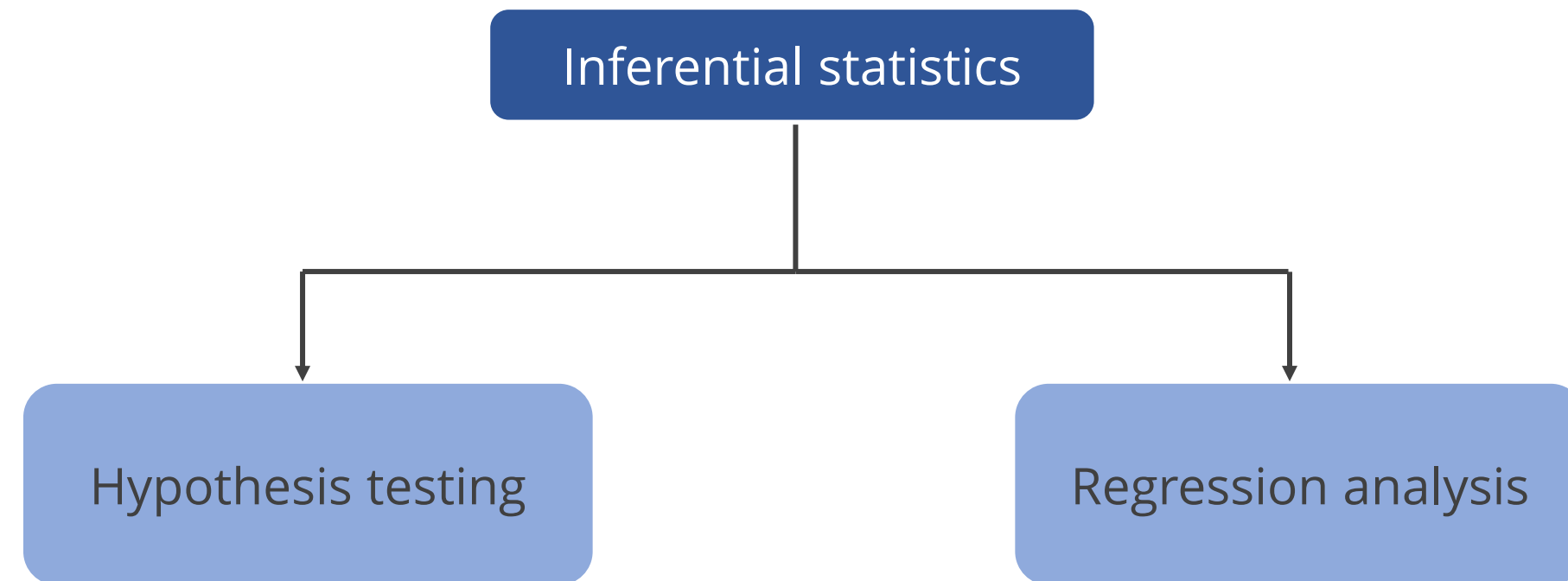
Inferential Statistics

It is commonly used in:



Types of Inferential Statistics

Inferential statistics can be broadly classified into two main categories:

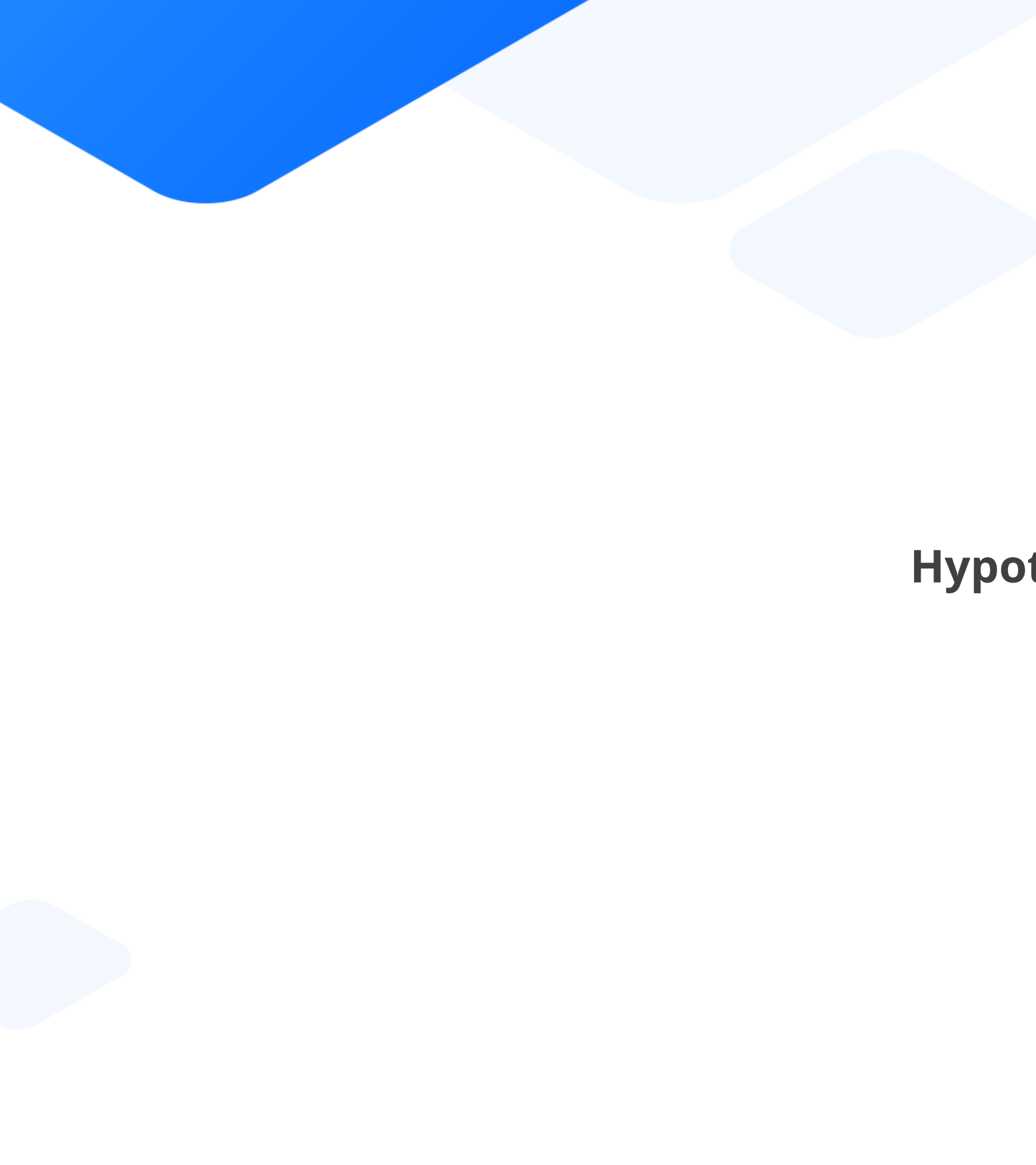


Discussion: Inferential Statistics

Duration: 15 minutes



- What is inferential statistics?
Answer: It refers to a branch of statistics that focuses on making inferences or predictions about a population using sample data.
- What are the different types of inferential statistics?
Answer: Hypothesis testing and regression analysis are two distinct types of inferential statistics.



Hypothesis

Discussion: Hypothesis Testing

Duration: 15 minutes



- What is a hypothesis?
- What is a p-value?

Hypothesis

A hypothesis is a predictive statement that can be tested using scientific methods and establishes a relationship between an independent variable and a dependent variable.



It provides a prediction about the relationship between variables or the distribution of a population.

Hypothesis

Hypotheses are typically formulated based on:

Prior knowledge

Observations

Assumptions

The hypotheses are tested using statistical methods.

Hypothesis

Some examples of hypothesis are given below:



The average lead time taken to procure materials from the supplier is three days.



The exercise regimen reduces an individual's weight by at least two pounds on average over a period of ten days.



The average hourly output of machine A is significantly higher than that of machine B.

Types of Hypothesis

The two types of hypothesis are null hypothesis and alternate hypothesis.

Null hypothesis	Alternative hypothesis
• It is a statement where there is no significant difference, relationship, or effect in the population or between two variables. • It is denoted as H_0. • It is usually the opposite of the research hypothesis, and the researcher tries to disprove it. • It usually includes equality.	• It represents a statement that contradicts the null hypothesis. • It is denoted as H_1 or H_a. • It corresponds to the research hypothesis, and the researcher tries to prove it. • It does not include equality.

Null and Alternative Hypothesis

Consider the following example:

Null Hypothesis (H_0): There is no significant difference in the average test scores between group A and group B.

Alternative Hypothesis (H_1): There is a significant difference in the average test scores between group A and group B.

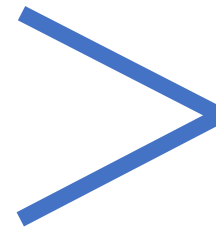
In this example, the null hypothesis assumes no difference between the two groups being tested, while the alternate hypothesis assumes a significant difference between them.

Hypothesis

Consider the following example of sales:



The average monthly sales (μ)



500 units owing to the sales promotion drive

Examples of Hypothesis

Assume null and alternative hypothesis as follows:

Null hypothesis: H_0

- The sales promotion drive has an insignificant impact on the average monthly sales.
- The historical average (μ_0) holds $\mu = \mu_0$.

Alternative hypothesis: H_1

- The sales promotion drive has a significant impact on the average monthly sales.
- $\mu > \mu_0$

Examples of Hypothesis

Consider the hourly output of two machines:

Null hypothesis: H_0

The average hourly output of machine A (μ_1) differs insignificantly from machine B (μ_2).

$$\mu_1 \approx \mu_2$$

Alternative hypothesis: H_1

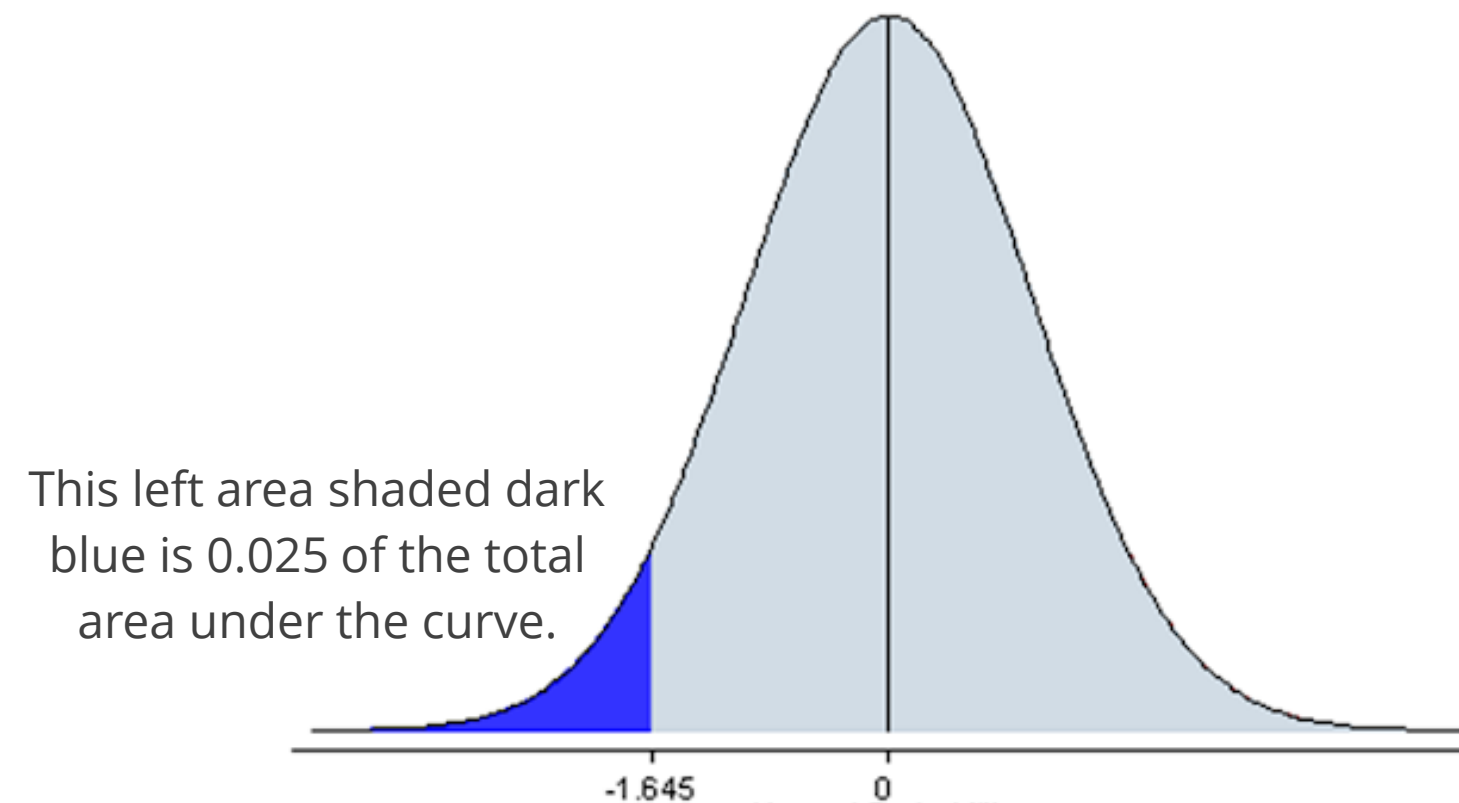
The average hourly output of machine A (μ_1) is significantly larger than that of machine B (μ_2).

$$\mu_1 > \mu_2$$

The null hypothesis is generally the negation of the assertion, while the alternative hypothesis is the assertion itself.

One-Tailed Tests

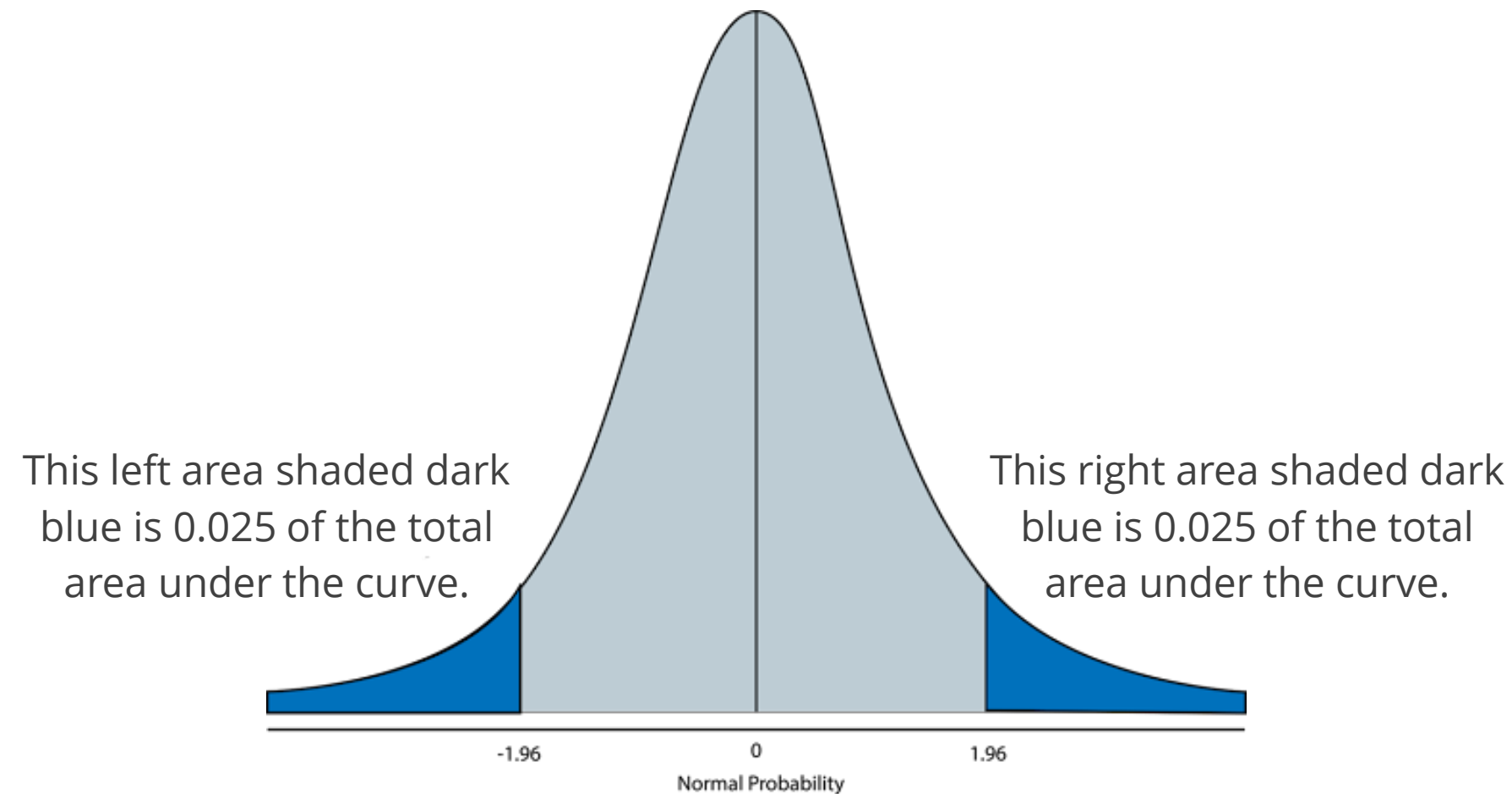
It is a test where the hypothesis being tested predicts the direction of the difference or relationship between variables.



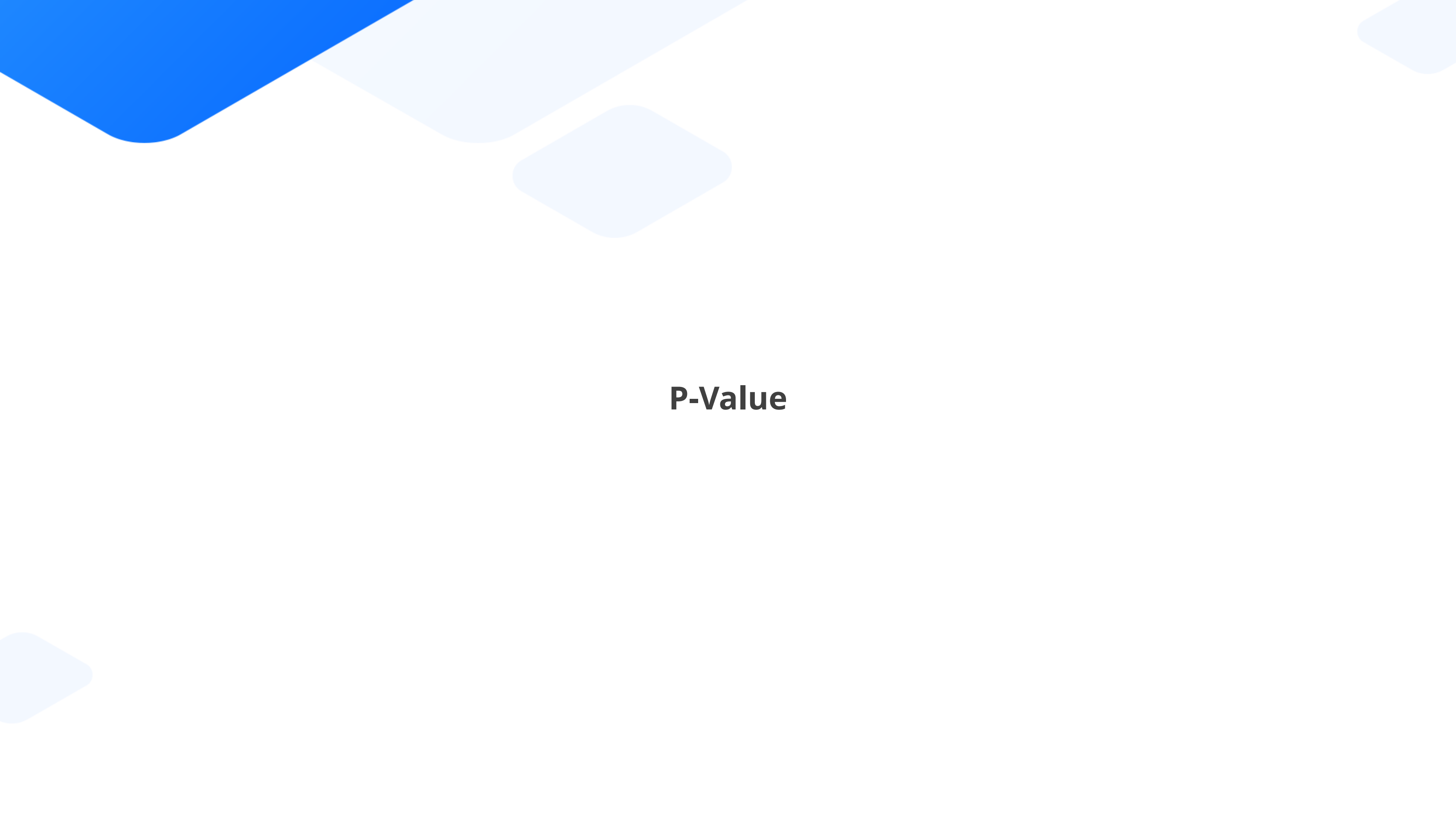
In this test, the null hypothesis states that there is no difference or relationship between the variables, while the alternative hypothesis specifies the direction of the difference or relationship.

Two-Tailed Tests

A two-tailed test checks for differences in either direction (positive or negative) between a sample and population.



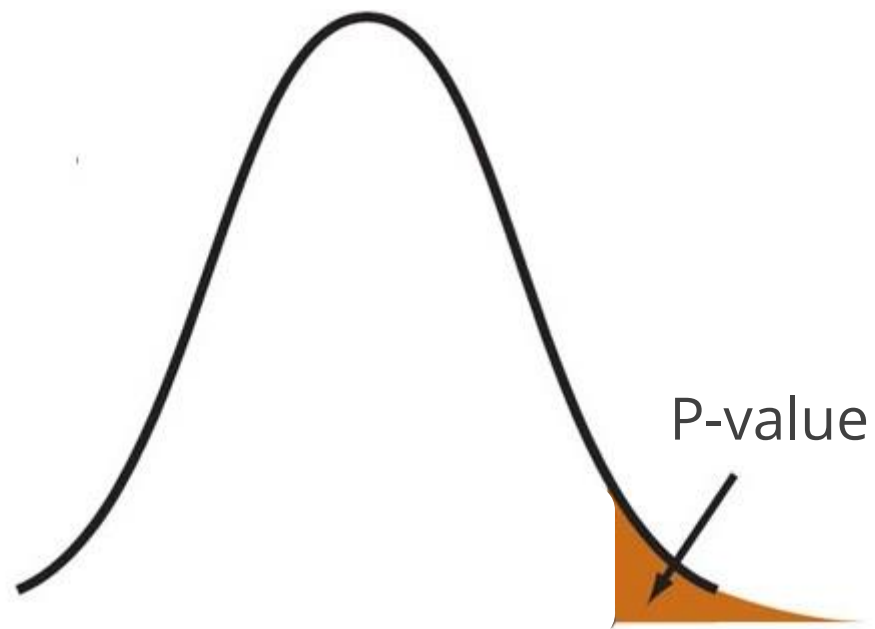
In this test, the null hypothesis states that there is no significant difference between the sample and the population, while the alternative hypothesis states that there is a significant difference in either direction.



P-Value

P-Value

It is defined as the probability of obtaining a result that is as extreme or more extreme than the observed data.



- The p-value is a measure of marginal significance in hypothesis testing and represents the probability of the observed event occurring.
- It serves as an alternative to the rejection point and indicates the minimum level of significance at which the null hypothesis would be rejected.
- A small p-value provides stronger evidence in favor of the alternative hypothesis.

P-Value

P-values are usually automatically calculated using statistical software.

The p-value is calculated based on the null distribution of the test statistic.

In case the test statistic is far from the mean of the null distribution, the p-value obtained is small.

It indicates that the test statistic is unlikely to have occurred under the null hypothesis.

P-Value

The formula for calculating the p-value is:

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$$

Where:

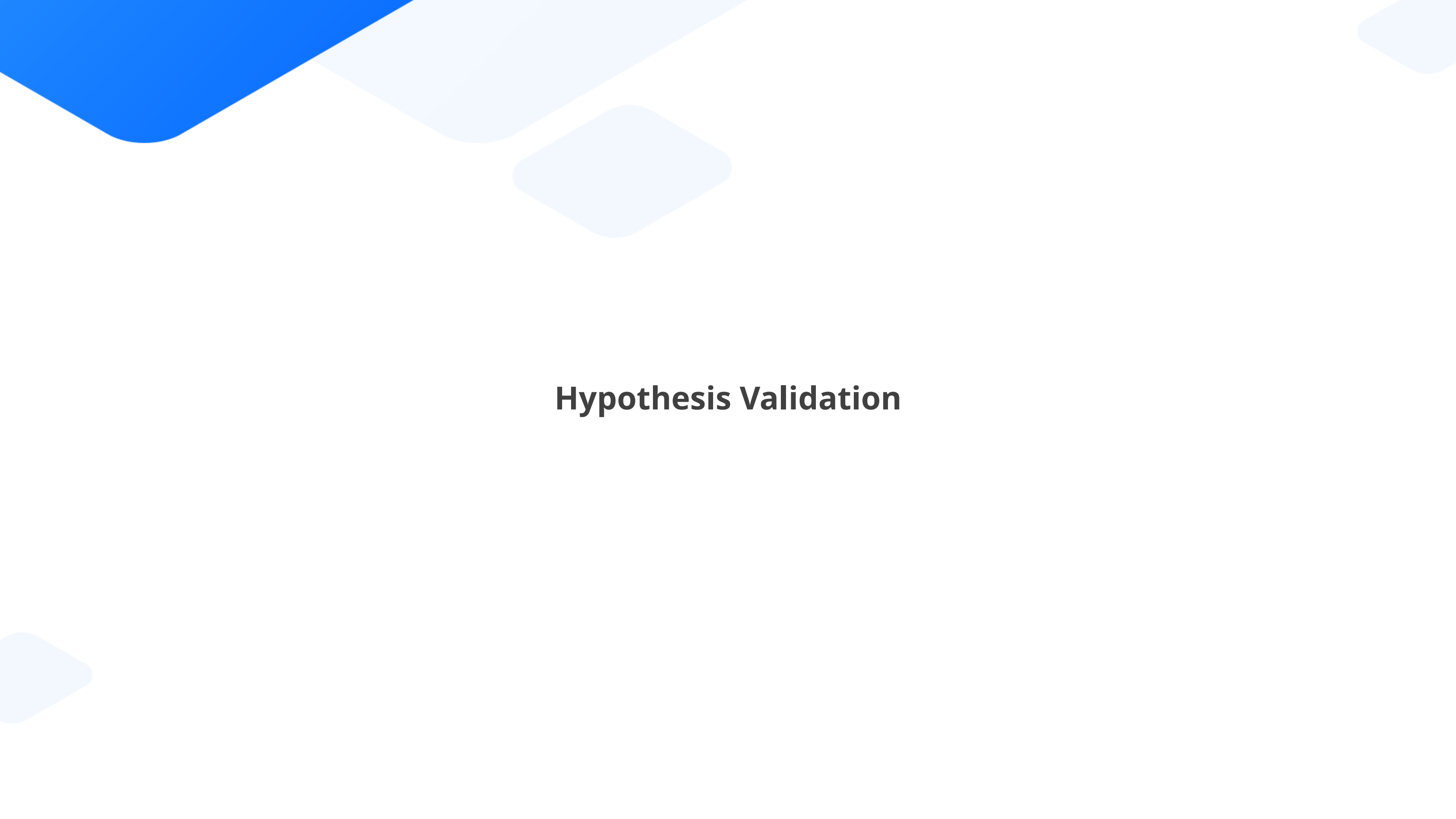
- $\hat{p}$ is the sample proportion
- p_0 is the assumed population proportion in the null hypothesis
- n is the sample size

P-Values and Statistical Significance

The level of statistical significance is represented as a p-value between 0 and 1.

The p-value table shows the hypothesis interpretations:

P-value	Decisions
P-value > 0.05	The result is not statistically significant and thus does not reject the null hypothesis.
P-value < 0.05	The result is statistically significant and generally rejects the null hypothesis in favor of the alternative hypothesis.
P-value < 0.01	The result is highly statistically significant and thus rejects the null hypothesis in favor of the alternative hypothesis.



Hypothesis Validation

Validation of Hypothesis

Validation tests the model or system with an independent dataset or real-world scenarios to assess performance, adherence to metrics, and reliability.



It involves testing the validity of a hypothesis through statistical analysis and evidence.

It helps researchers make informed decisions by either accepting or rejecting the null hypothesis based on the results of the analysis.

Validation of Hypothesis

The process of hypothesis validation typically involves the following steps:

Feature	Hypothesis testing	Validation
Purpose	To assess whether there is a statistically significant difference between two groups or variables	To assess the accuracy of a model or hypothesis
When to use	Early in the research process	Later in the research process
Steps involved	• State the hypothesis • Test the hypothesis	• Build the model • Validate the model • Improve the model



Hypothesis Testing

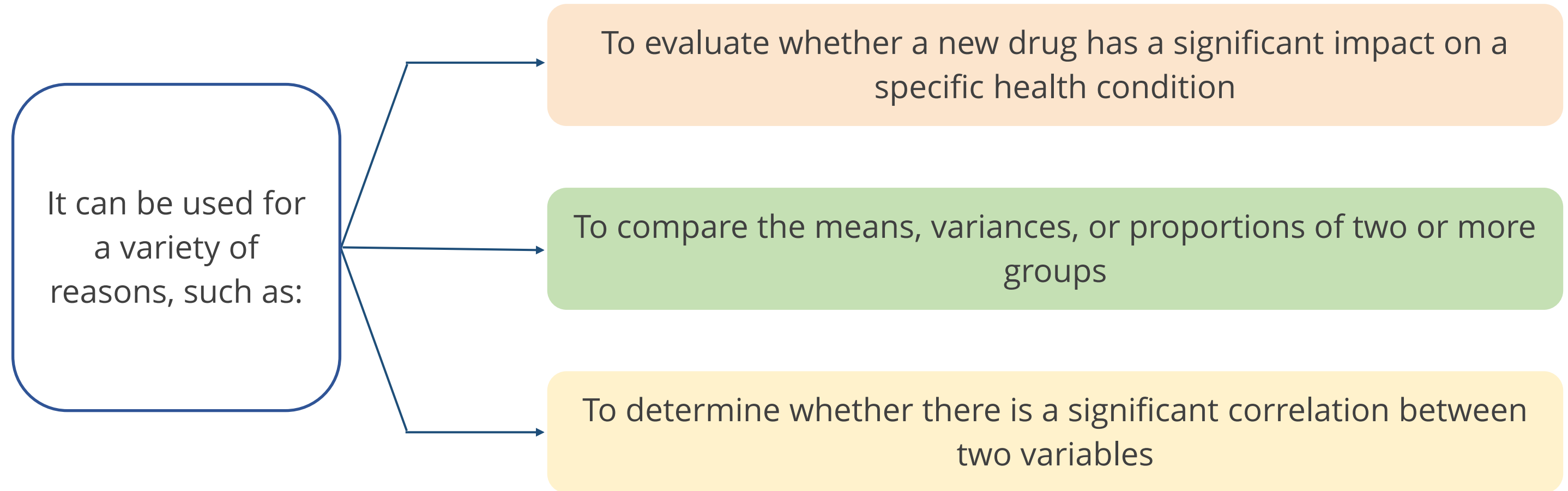
Hypothesis Testing

It is a statistical method used to determine whether a certain statement about a population is likely to be true or not.



It estimates the relationship between two statistical variables.

Hypothesis Testing



Benefits of Hypothesis Testing

1

It enables the evaluation of the strength of a claim or assumption before applying it to the dataset.

2

It provides a reliable framework for making data-driven decisions regarding the population of interest.

3

It assists researchers in effectively extrapolating data from a sample to a larger population.

4

It measures the validity and reliability of outcomes in a systematic investigation.

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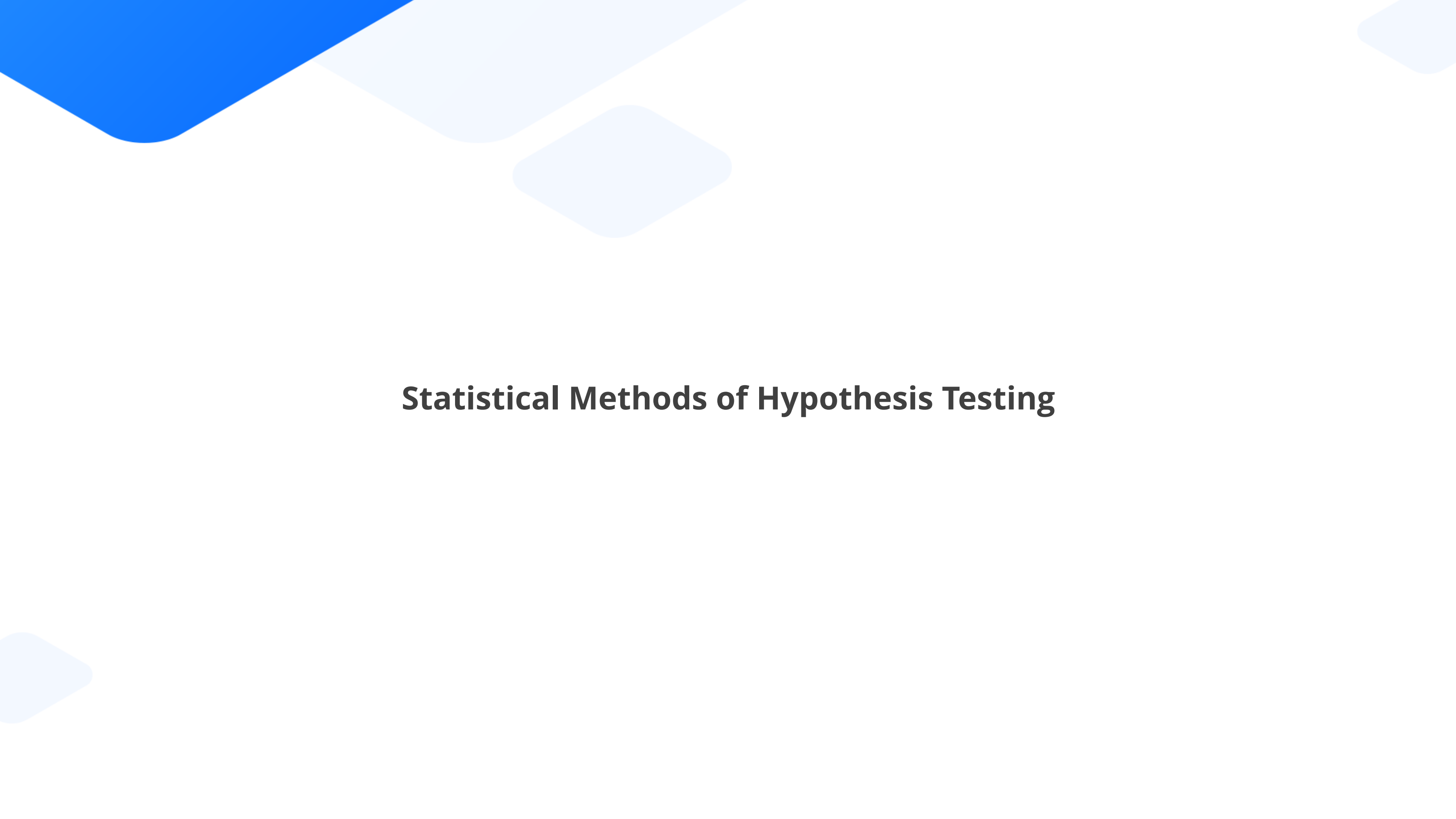
It helps researchers determine whether the data from the sample is statistically significant.

Discussion: Hypothesis Testing

Duration: 15 minutes



- What is a hypothesis?
Answer: A hypothesis is a predictive statement that can be tested using scientific methods and establishes a relationship between an independent variable and a dependent variable.
- What is a p-value?
Answer: It is the probability of getting a result that is either the same or more extreme than the actual observations.



Statistical Methods of Hypothesis Testing

Discussion: Statistical Methods of Hypothesis Testing

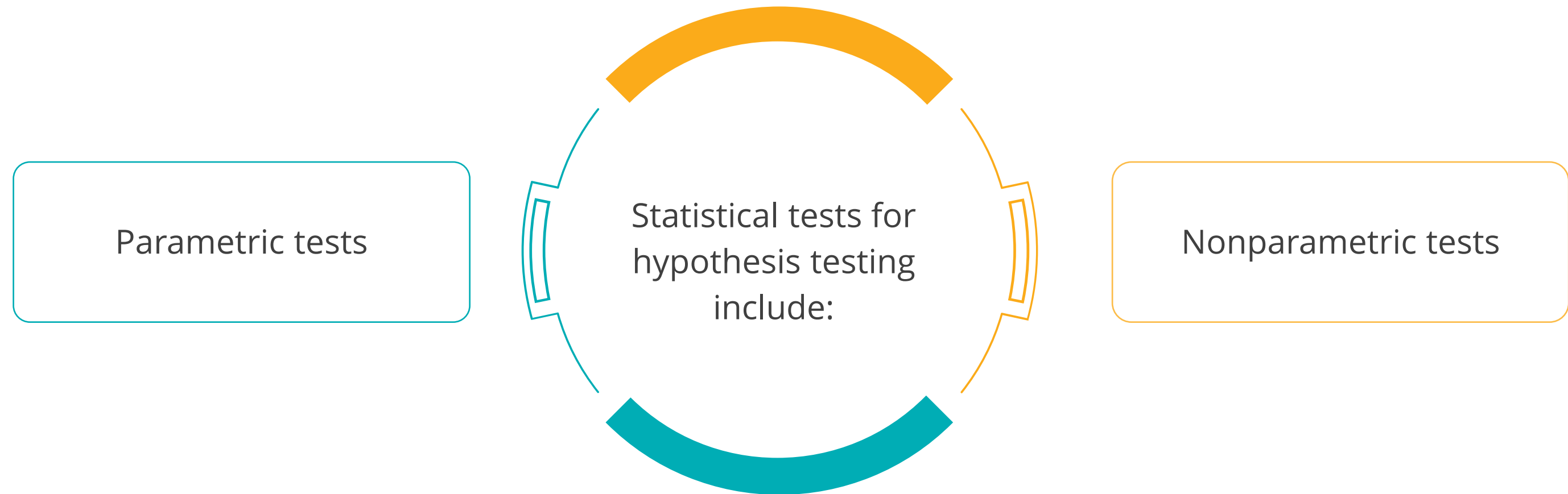
Duration: 15 minutes

- What is a parametric test?
- What is a z-test, and what are its types?



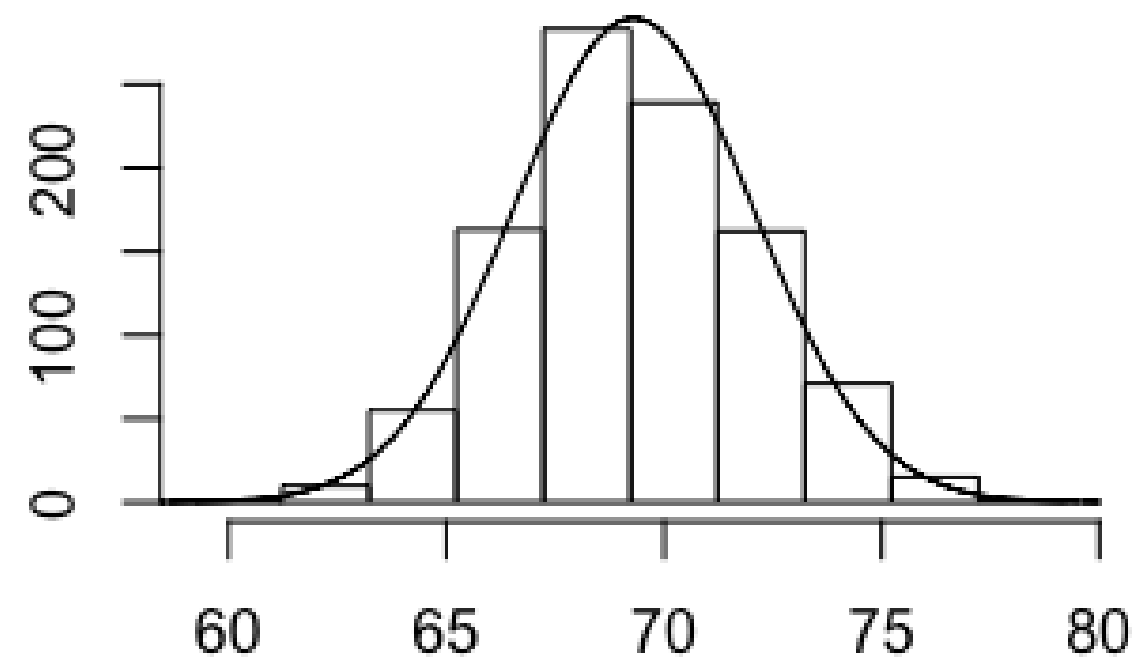
Statistical Methods of Hypothesis Testing

Statistical tests are used in hypothesis testing to check whether the null hypothesis is rejected or not.



Parametric Tests

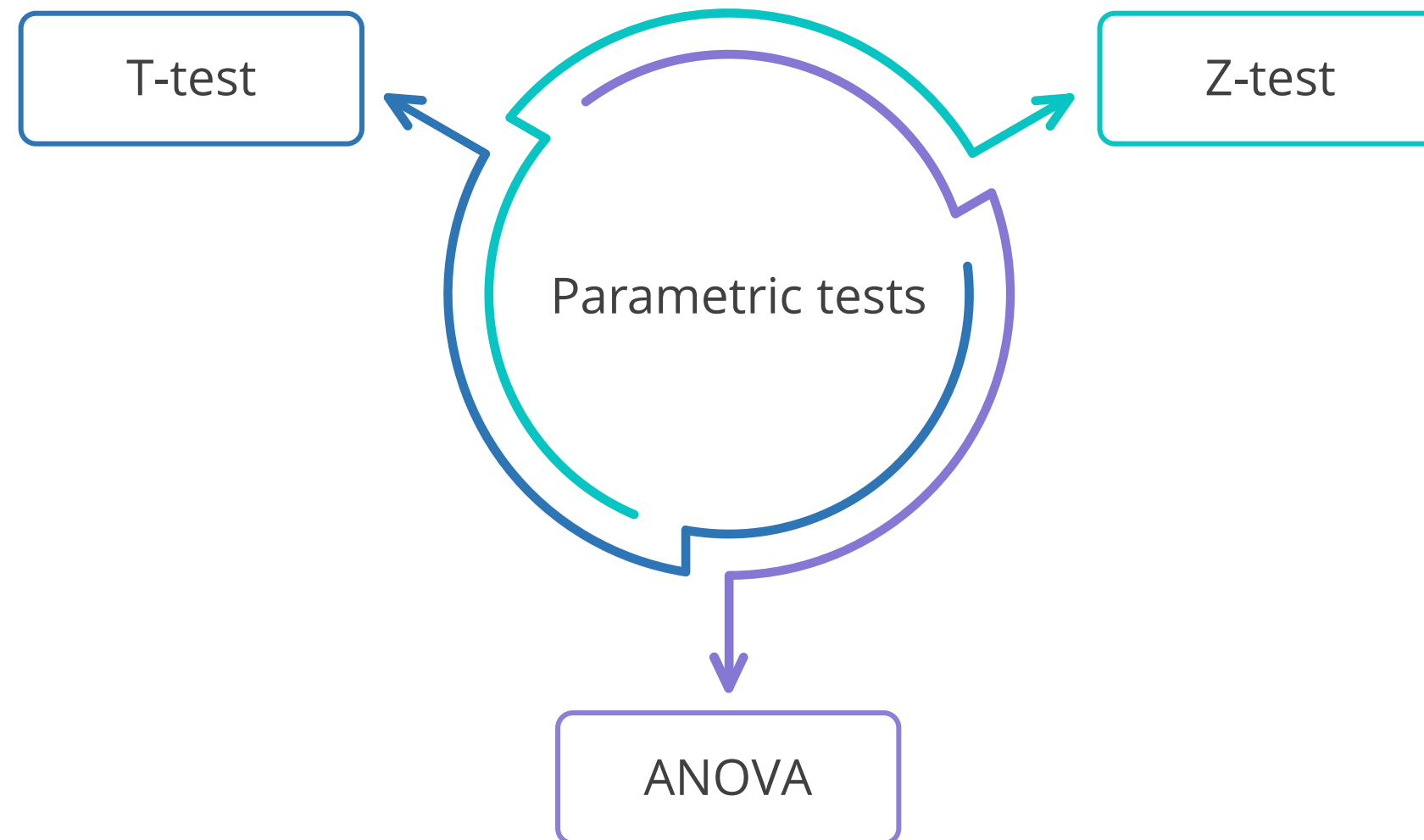
A parametric test is a test in which parameters are assumed and the population distribution is always known.



These tests require the data to meet certain assumptions, such as normality and equal variances.

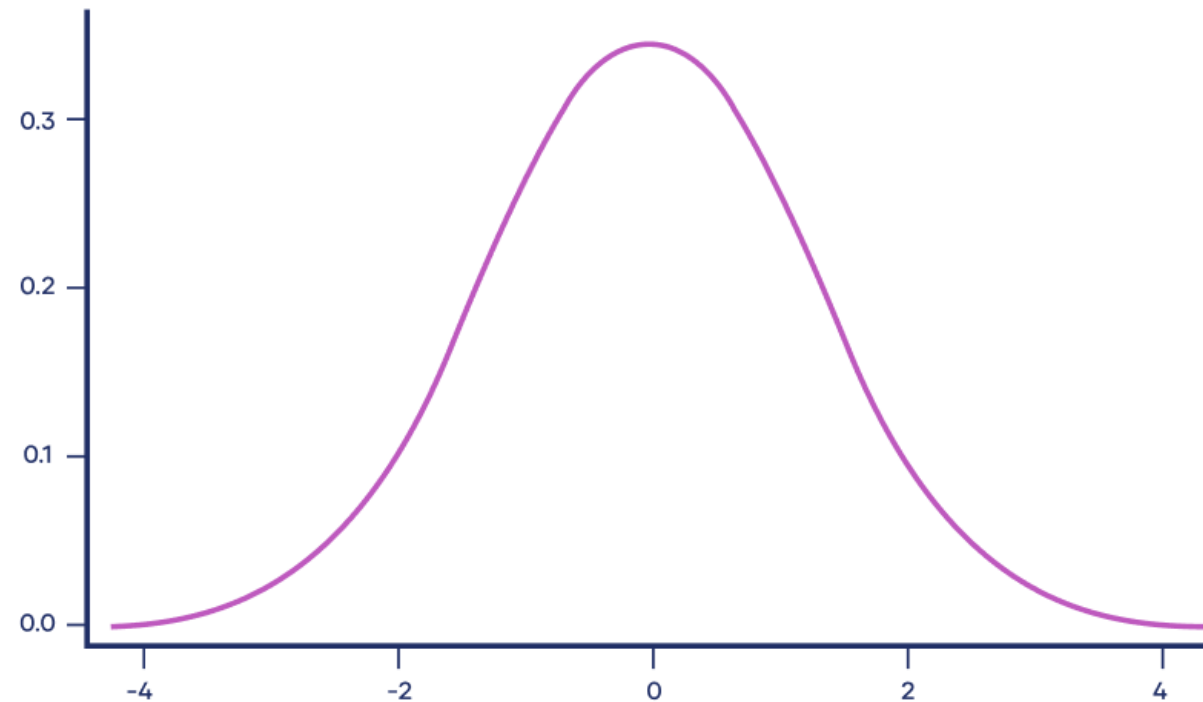
Types of Parametric Tests

The most widely used parametric tests are:



T-Test

The t-test is a commonly used inferential statistics technique that helps determine whether there is a significant difference between the means of two groups of data.



- It is used to make inferences about population parameters based on sample data.
- It involves comparing the means of two groups and calculating the t-value, which is a ratio of the difference between the means and the variability within the group.
- It determines if the calculated t-value is large enough to indicate a statistically significant difference between the two means.

One-Sample T-Test

It compares the mean of a population $\bar{x}$ from n samples with a specified theoretical mean μ .

The t-test formula is:

$$t = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Where:

- $\bar{x}$ is the mean of the sample
- μ is the assumed mean
- σ is the standard deviation
- n is the number of observations

Independent Sample T-Test

It compares the means of two sample groups to determine if there is a significant statistical difference between them.

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$$

Where:

- t = Student's t-test
- $\bar{x}_1$ = Mean of group 1
- $\bar{x}_2$ = Mean of group 2
- s_1 = Standard deviation of group 1
- s_2 = Standard deviation of group 2
- n_1 = Number of observations in group 1
- n_2 = Number of observations in group 2

Paired Sample T-Test

It compares the means of two measurements taken from the same individual, object, or related units.

$$t = \frac{\Sigma(x_1 - x_2)}{\frac{s}{\sqrt{n}}}$$

- t = Student's t-test
- $x_1 - x_2$ = Mean difference of the pairs
- s = Standard deviation
- n = Sample size

T-Test: Example

A researcher wants to determine whether there is a significant difference in the weight of newborn babies between two hospitals in a city. The researcher randomly selects 20 newborns from hospital A and 20 newborns from hospital B, recording their weights in pounds. The mean weight for the group from hospital A is 7.5 with a standard deviation of 0.8. For the group from hospital B, the mean weight is 7.1 with a standard deviation of 1.2. Can we conclude that there is a significant difference between the weights of newborns in the two hospitals?

This is an independent sample's t-test problem since the two groups being compared are independent of each other.

T-Test: Example

Using the formula for t-value:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$$

$$t = (7.5 - 7.1) / (\text{sqrt}((0.8^2/20) + (1.2^2/20)))$$

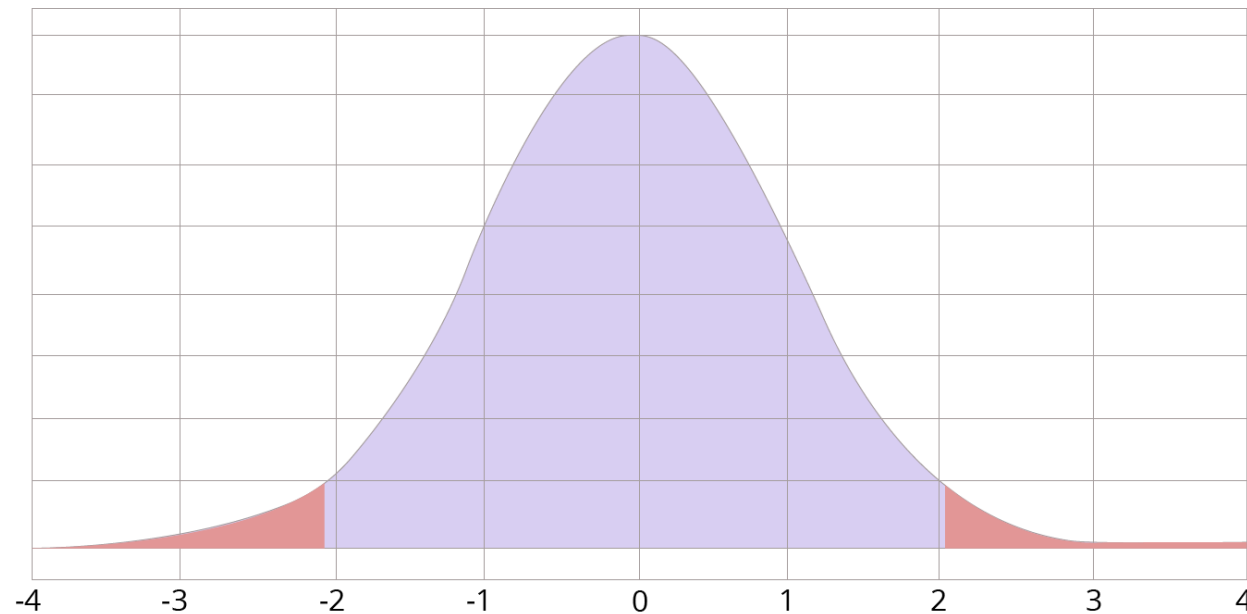
$$t = 1.235$$

Assuming a significance level of 0.05 and df = 38, the critical t-value is 2.024.

Since the calculated t-value of 1.235 is less than the critical t-value of 2.024, it cannot be concluded that there is a significant difference in the weight of newborn babies between the two hospitals in the city.

Z-Test

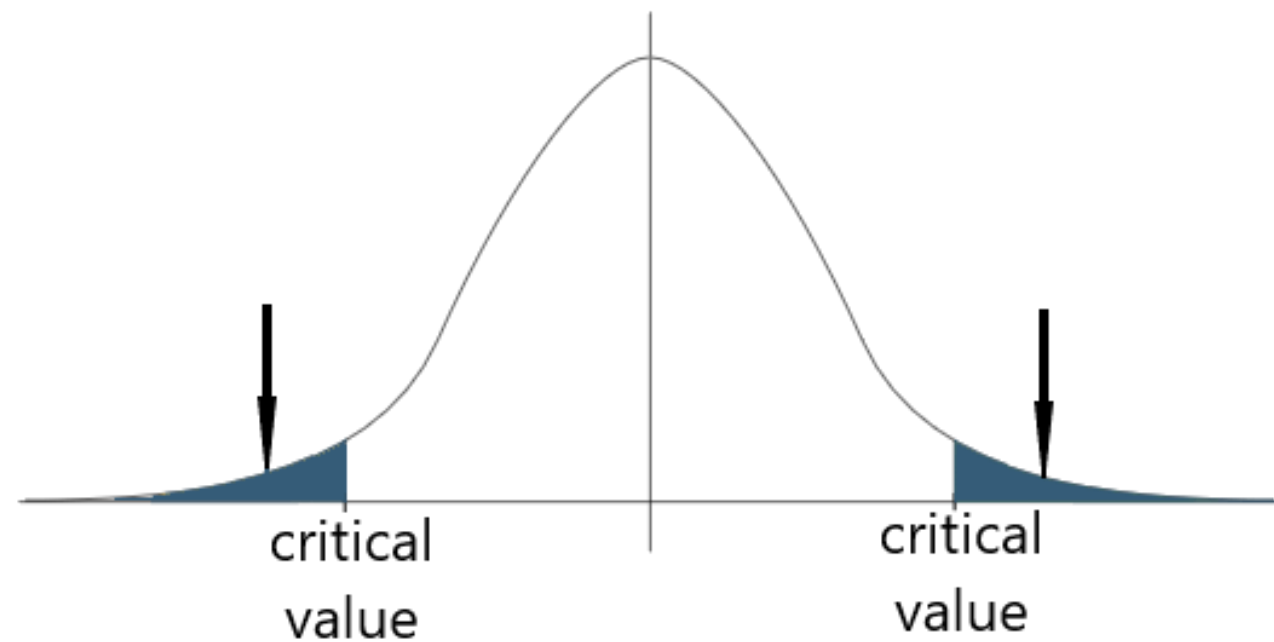
It checks if the means of two populations are different or not, provided the data follows a normal distribution.



- It can be performed on one sample, two samples, or on proportions for hypothesis testing.
- It checks if the means of two large samples are different or not when the population variance is known.
- It can be classified into left-tailed, right-tailed, and two-tailed hypothesis tests depending on the parameters of the data.

Critical Value

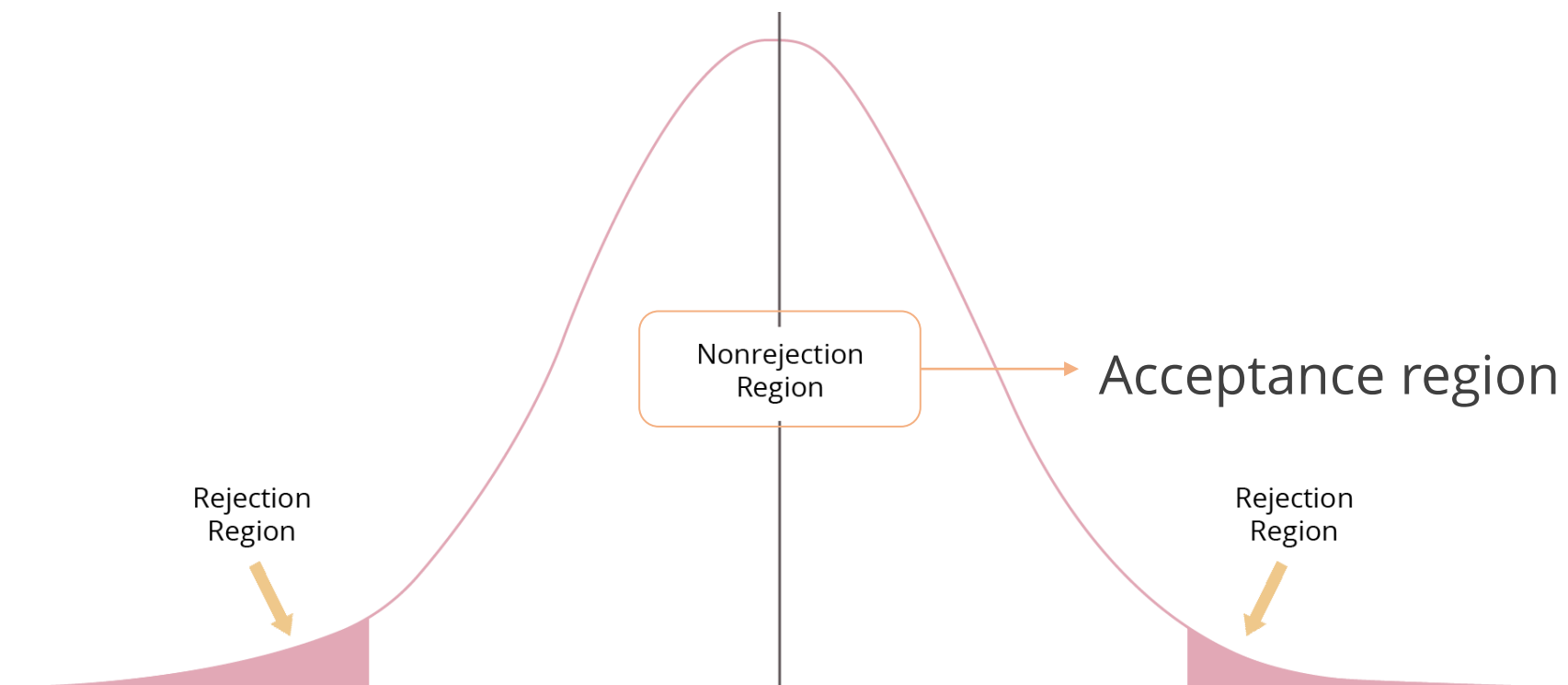
It is a value that is compared to a test statistic to determine whether the null hypothesis is to be rejected or not.



- If the value of the test statistic is less extreme than the critical value, then the null hypothesis cannot be rejected.
- If the test statistic is more extreme than the critical value, the null hypothesis is rejected, and the alternative hypothesis is accepted.

Critical Value

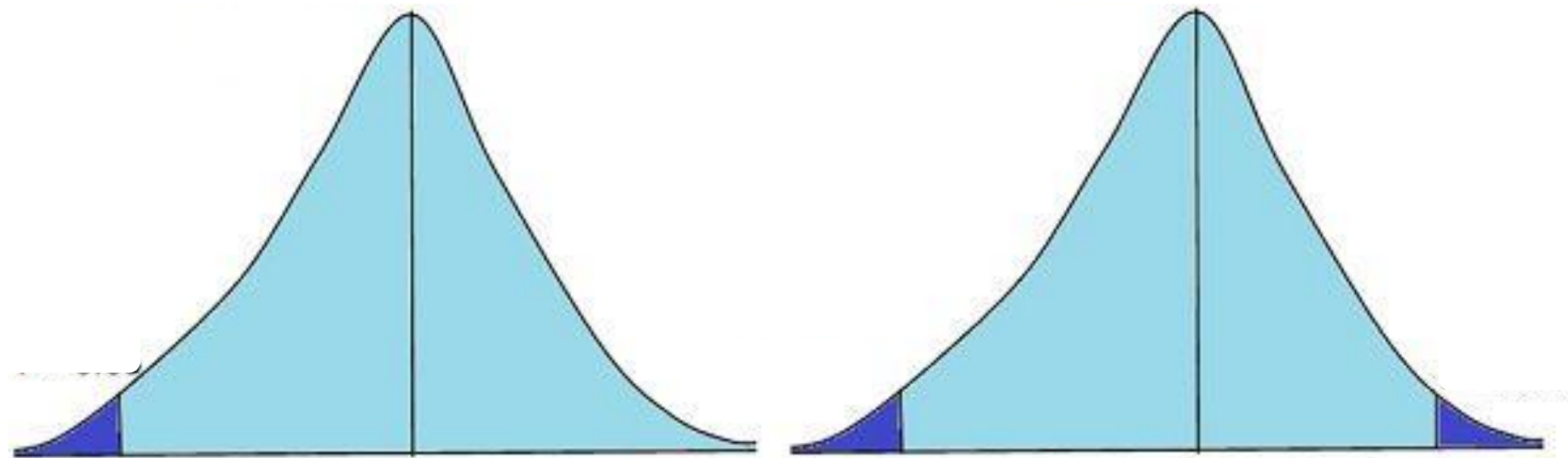
The critical value divides the distribution graph into acceptance and rejection regions.



If the test statistic is in the rejection region, reject the null hypothesis; otherwise, do not reject it.

Critical Value Confidence Interval

The confidence interval can compute the critical values for a one-tailed or two-tailed test.



A confidence interval is the probability that a population parameter will fall between a set of values for a certain proportion of times.

Critical Value Confidence Interval

Suppose a confidence interval of 93% has been specified for conducting a hypothesis test

The critical value can be determined as follows:

Step 1: Subtract the confidence level from 100%. $(100\% - 93\%) = 7\%$.

Step 2: Convert this value to decimals to get α . Thus, $\alpha = 0.07\%$.

Step 3: For a one-tailed test, the alpha level remains the same as in step 2. However, for a two-tailed test, the alpha level is divided by 2.

Step 4: Look up the critical value from the distribution table based on the test type and alpha value.

One-Sample Z-Test

It compares the sample mean with the population mean when the population standard deviation is known to determine if there is a difference.

The formula for one sample z-test is given as follows:

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Where:

- $\bar{x}$ is the sample mean
- μ is the population mean
- σ is the population standard deviation
- n is the sample size

One-Sample Z-Test

The algorithm for conducting a one-sample z-test using the z-test statistic is as follows:

Measure	Left-tailed test	Right-tailed test	Two-tailed test
Null hypothesis	$H_0 : \mu = \mu_0$	$H_0 : \mu = \mu_0$	$H_0 : \mu = \mu_0$
Alternative hypothesis	$H_1 : \mu < \mu_0$	$H_1 : \mu > \mu_0$	$H_1 : \mu \neq \mu_0$
Decision criteria	If the z statistic < z critical value, then reject the null hypothesis	If the z statistic > z critical value, then reject the null hypothesis	If the z statistic > z critical value, then reject the null hypothesis

Two-Sample Z-Test

It checks if there is a difference between the means of two samples.

The formula for a two-sample z-test is given as follows:

$$\frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}$$

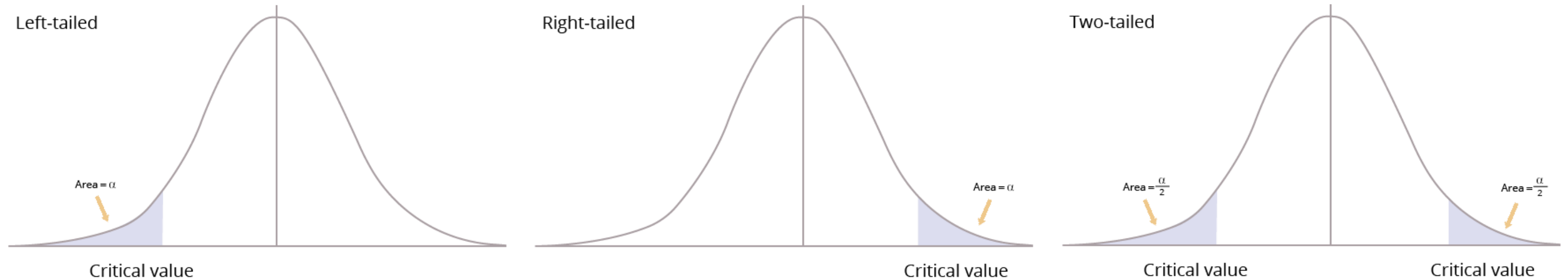
Where:

- $\bar{x}_1 - \bar{x}_2$ are sample means
- σ_1 and σ_2 are population standard deviations
- n_1 and n_2 are sample sizes

Two-Sample Z-Test

The two-sample z-test can be set up in the same way as the one-sample test, and it will be used to compare the means of the two samples.

Example: The null hypothesis is given as $H_0 : \mu_1 = \mu_2$



Z-Test: Example

Problem Statement: An online medicine shop claims a mean delivery time of less than 120 minutes, with a standard deviation of 30 minutes. Is there enough evidence to support this claim at a significance level of 0.05, based on 49 examined orders with a mean delivery time of 100 minutes?

Solution: Given a sample size of 49 and a known population standard deviation, this scenario corresponds to a left-tailed one-sample z-test.

$$H_0 : \mu = 120$$

$$H_1 : \mu < 120$$

- The critical value α is at -1.645.
- A negative sign is used as this is a left-tailed test.

Z-Test: Example

Using the equation:

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

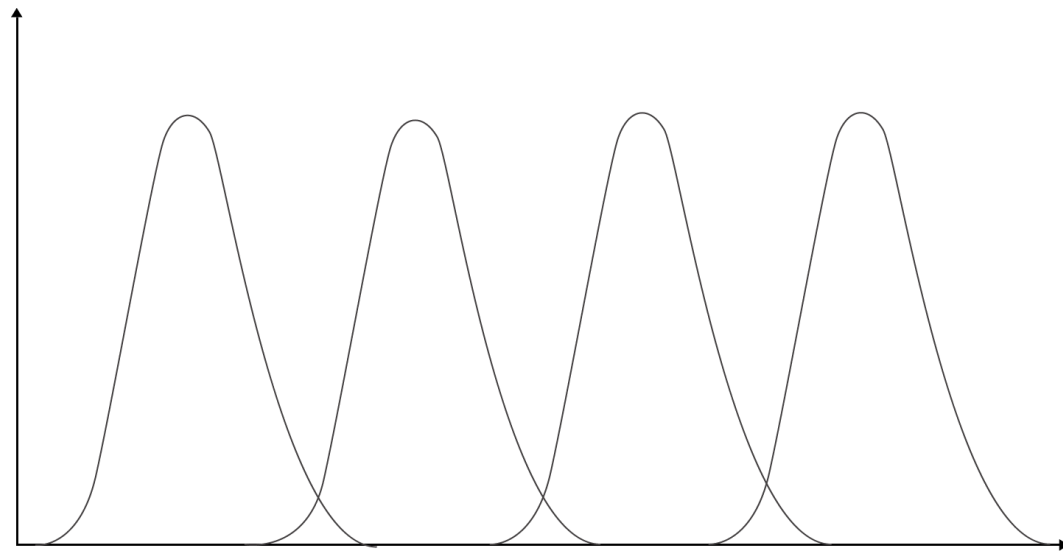
$$\bar{x} = 100, \mu = 120, n = 49, \sigma = 30$$

$$z = \frac{100 - 120}{30 / \sqrt{49}} = -4.66$$

As $-4.66 < -1.645$, the null hypothesis is rejected; it is concluded that there is enough evidence to support the medicine shop's claim.

Analysis of Variance (ANOVA)

ANOVA is a hypothesis test used to compare the means of two or more groups to determine if they are equal or not.



- It is used to determine if the null hypothesis can be rejected or not based on the statistical significance exhibited by the parameters.
- The decision is made by comparing the ANOVA test statistic with the critical value.
- It is a powerful technique for comparing the means of several populations.

Analysis of Variance (ANOVA)

The formula for ANOVA is:

$$F = \frac{MST}{MSE}$$

Where:

F = ANOVA coefficient

MST = Mean sum of squares due to treatment

MSE = Mean sum of squares due to error

One-Way ANOVA

It compares the means of two or more groups on a single independent variable.

It determines whether all the samples are the same.

It is used to determine whether there are any statistically significant differences between the means of three or more independent groups.

One-Way ANOVA

One-way ANOVA can be conducted to compare the average height of individuals from different countries, with the country serving as the independent variable and height as the dependent variable.



Two-Way ANOVA

It examines the effects of two independent variables on a single dependent variable.

It determines the effect of two variables on an outcome.

Some assumptions need to be considered when carrying out the two-way ANOVA test to ensure that the data conforms to these assumptions and increases the quality of the results.

Two-Way ANOVA

A two-way ANOVA compares the average height of individuals across different countries and genders, treating country and gender as independent variables and height as the dependent variable.



Discussion: Statistical Methods of Hypothesis Testing

Duration: 15 minutes

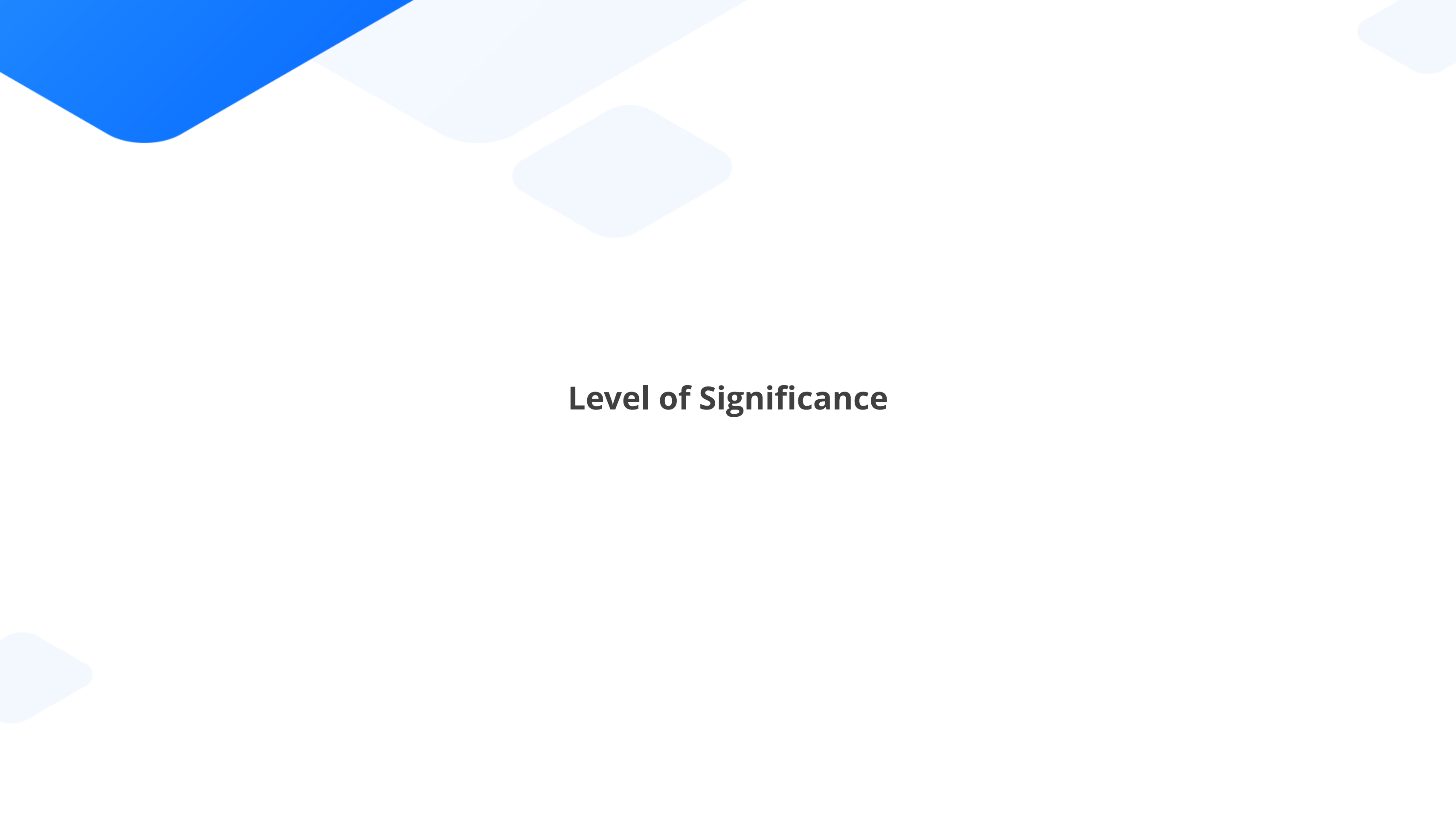


- What is a parametric test?

Answer: A parametric test assumes known population distribution and specific parameters. These tests have assumptions, including normality and equal variances, that must be met by the data.

- What is a z-test, and what are its types?

Answer: It is a test that is used to check if the means of two populations are different or not, provided the data follows a normal distribution. It can be classified into left-tailed, right-tailed, and two-tailed hypothesis tests depending on the parameters of the data. The different types of z-test are one-sample z-test and two-sample z-test.



Level of Significance

Discussion: Level of Significance

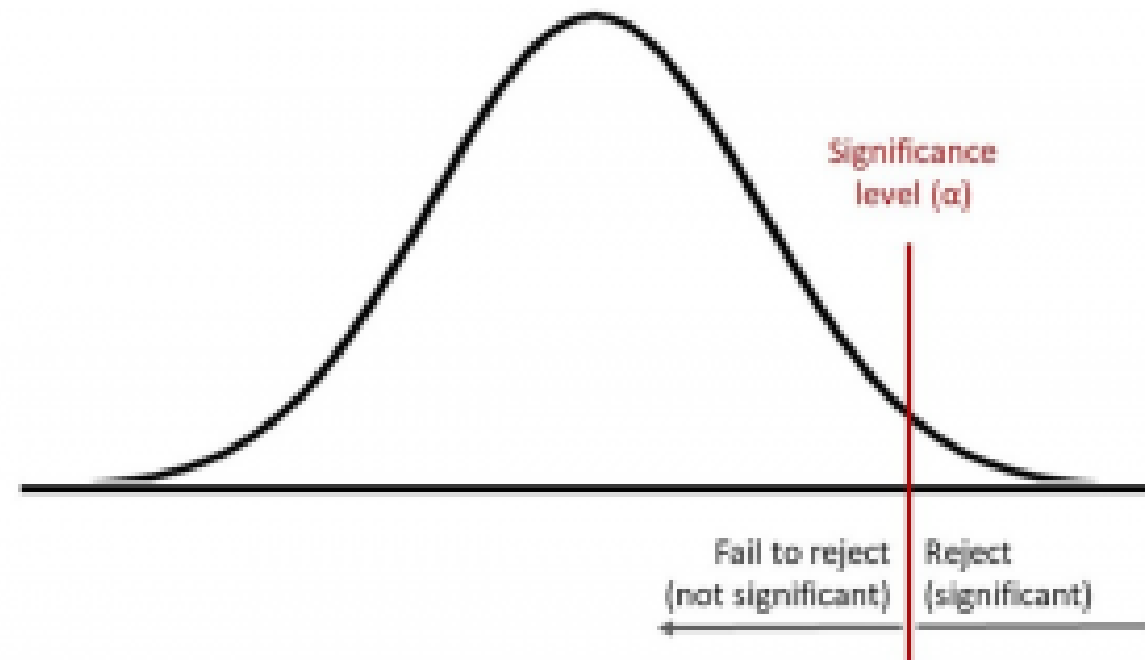
Duration: 15 minutes

- What is the level of significance?
- What are the outcomes of hypothesis testing?



Level of Significance

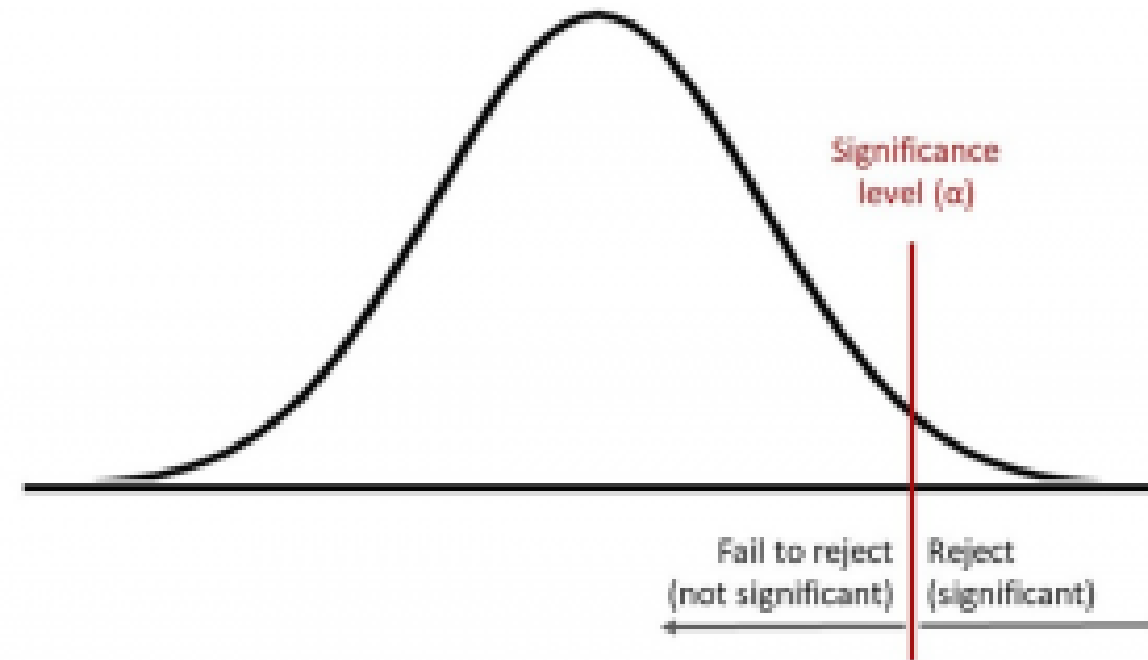
The level of significance, also known as the significance level, represents the probability of rejecting the null hypothesis when it is true.



It is denoted by the Greek letter alpha (α) and is typically set to 0.05 or 0.01.

Level of Significance

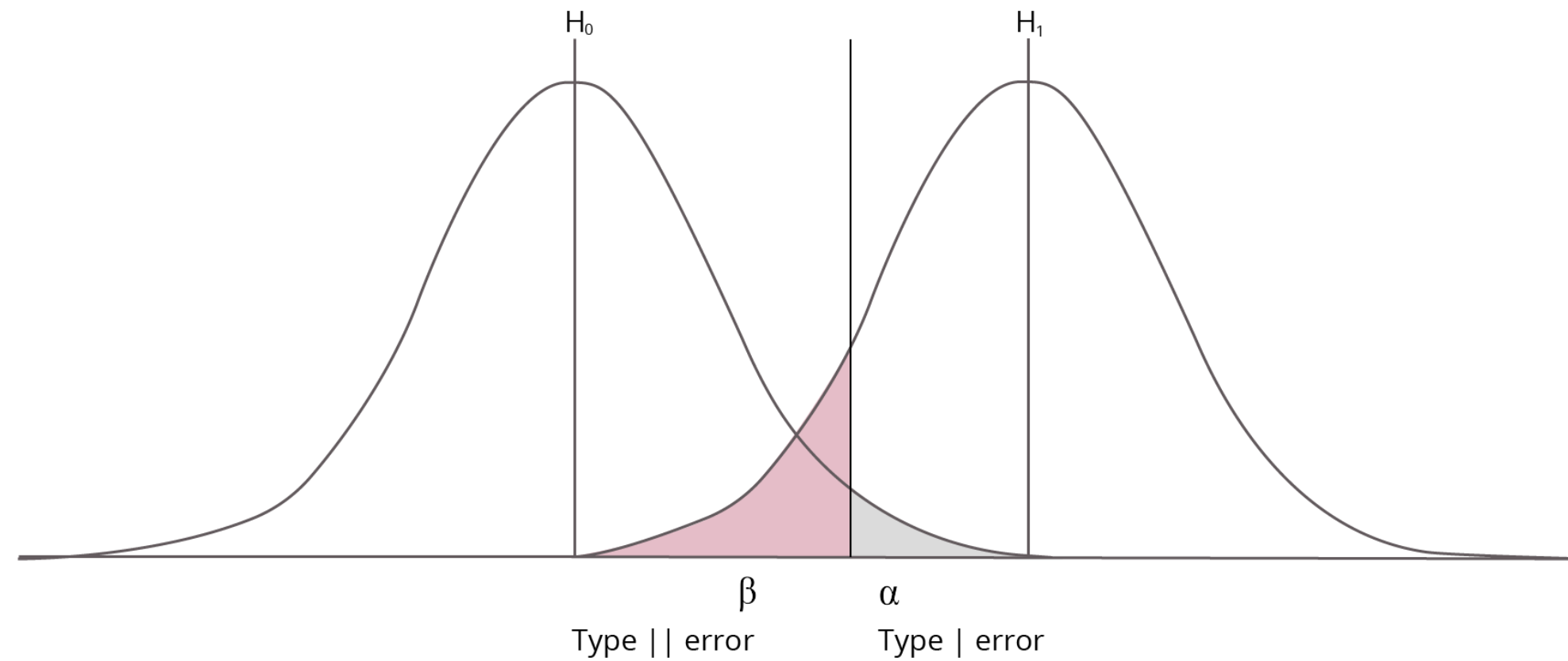
The level of significance plays a critical role in inferential statistics, as it guides researchers in determining whether to reject or fail to reject the null hypothesis.



It determines the cutoff value for the p-value.

Type I and Type II Errors

Type I and Type II errors can occur when making decisions based on statistical hypothesis testing.



Type I Error

A Type I error occurs when a researcher rejects a null hypothesis that is true.

This error is also known as a false positive.

It is more likely to occur when the level of significance is set too high, because it increases the chance of rejecting the null hypothesis even when it is true.

Example: If a researcher concludes that a new drug is effective when it is not, this would be a Type I error.

Type II Error

A Type II error occurs when a researcher fails to reject a null hypothesis that is false.

This error is also known as a false negative.

It is more likely to occur when the level of significance is set too low, because it increases the chance of failing to reject the null hypothesis even when it is false.

Example: If a researcher concludes that a new drug is not effective when it is, this would be a Type II error.

Level of Significance

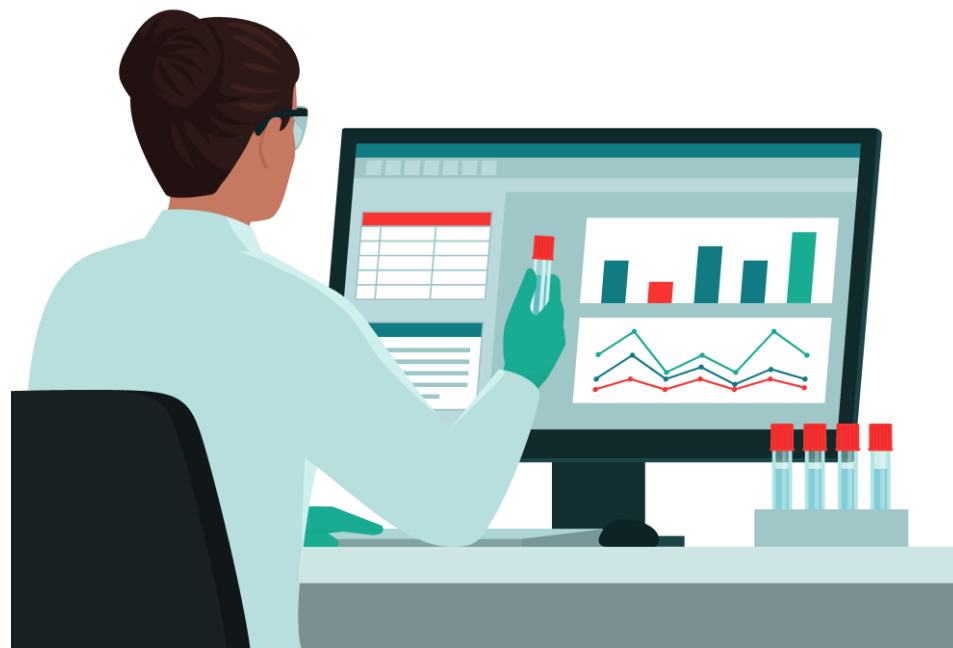
The level of significance provides a balance between the risk of making a type I error and the risk of making a type II error.



The researchers can determine the amount of evidence required to support the alternative hypothesis and decide whether to reject or not reject the null hypothesis.

Level of Significance: Example

Suppose a researcher is conducting a test to determine the effectiveness of a new drug in treating a specific medical condition.

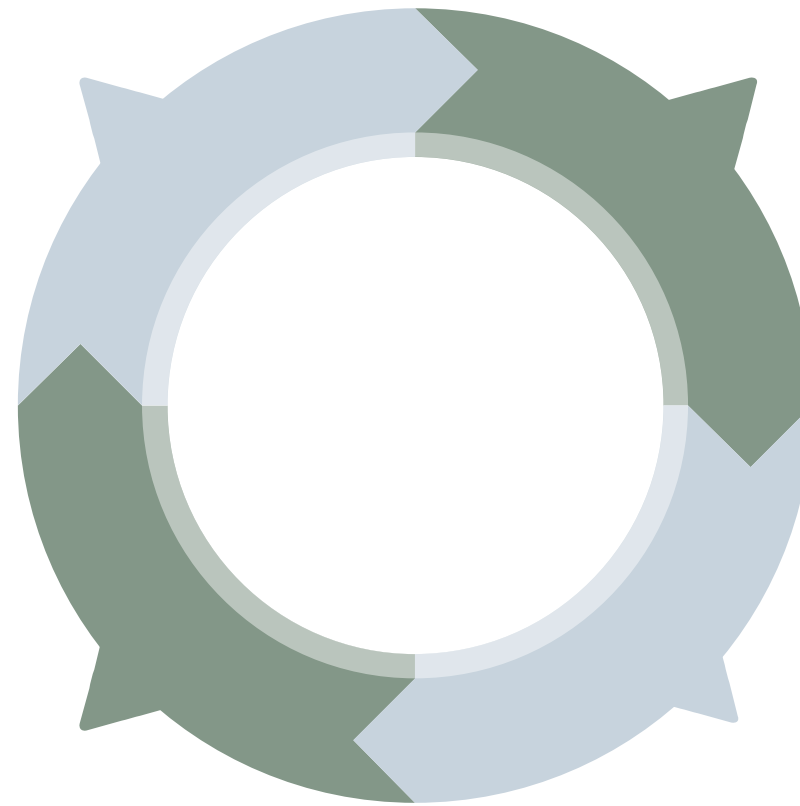


- If the researcher sets a level of significance of 0.05, it means that they are willing to accept a 5% chance of making a type I error and a 95% chance of making a correct decision.
- The level of significance helps ensure that the researcher makes informed decisions based on statistical evidence rather than relying on random chance or anecdotal observations.

Decision and Outcomes of Hypothesis Testing

H_0 (null hypothesis) is true, and it is rejected.

H_0 (null hypothesis) is true, and it is accepted.



H_0 (null hypothesis) is false, and it is accepted.

H_0 (null hypothesis) is false, and it is rejected.

Probability of Error

The probabilities of occurrence of type I and type II errors are denoted by:



Error Occurrences

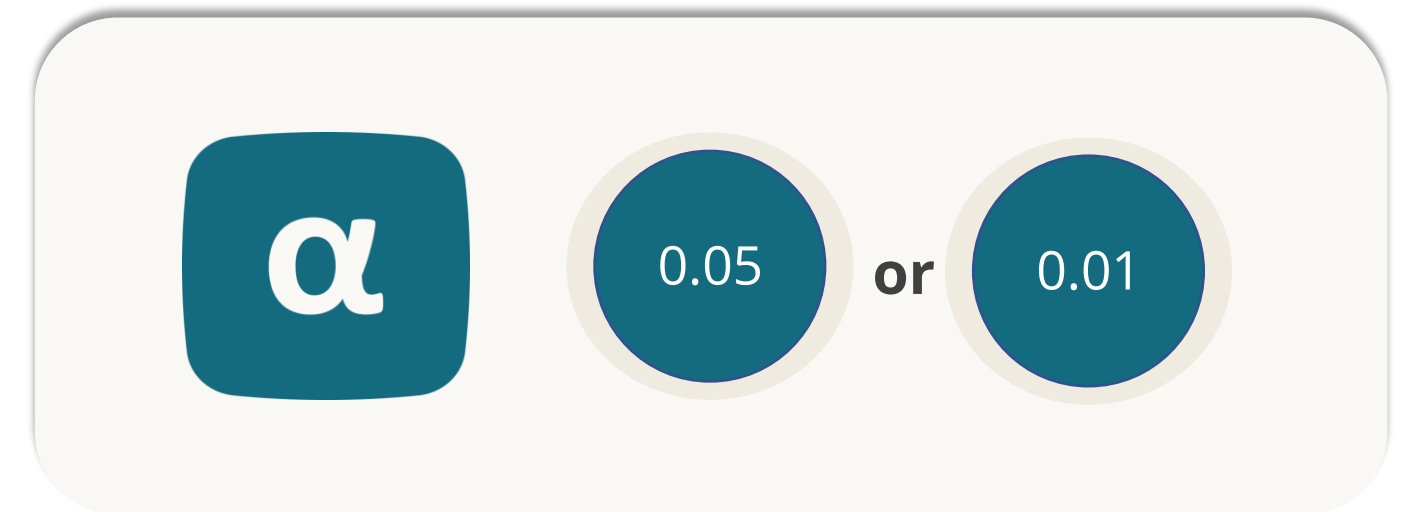
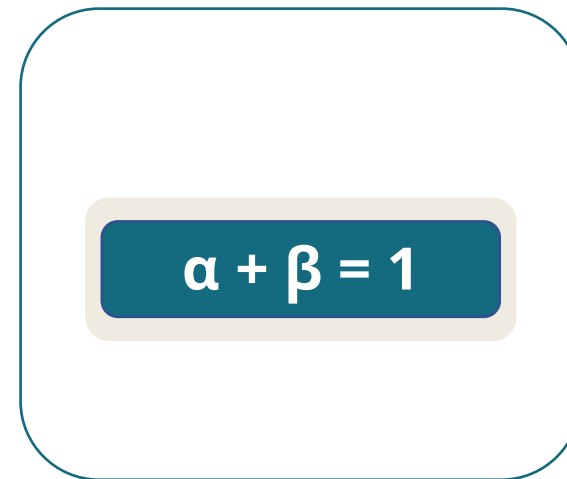
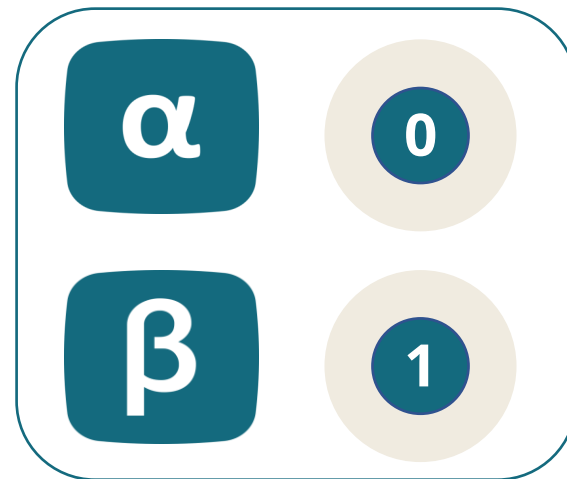
It is not possible to simultaneously make the two values of Alpha (α) and Beta (β) zero when making inferences based on samples.



This is possible with complete enumeration.

Error Occurrences

If one of them is set to zero, the other becomes one.



This does not mean that $\alpha + \beta = 1$.
(Alpha and Beta are equal to 1)

Large sample sizes are used so that the
Beta is also low.

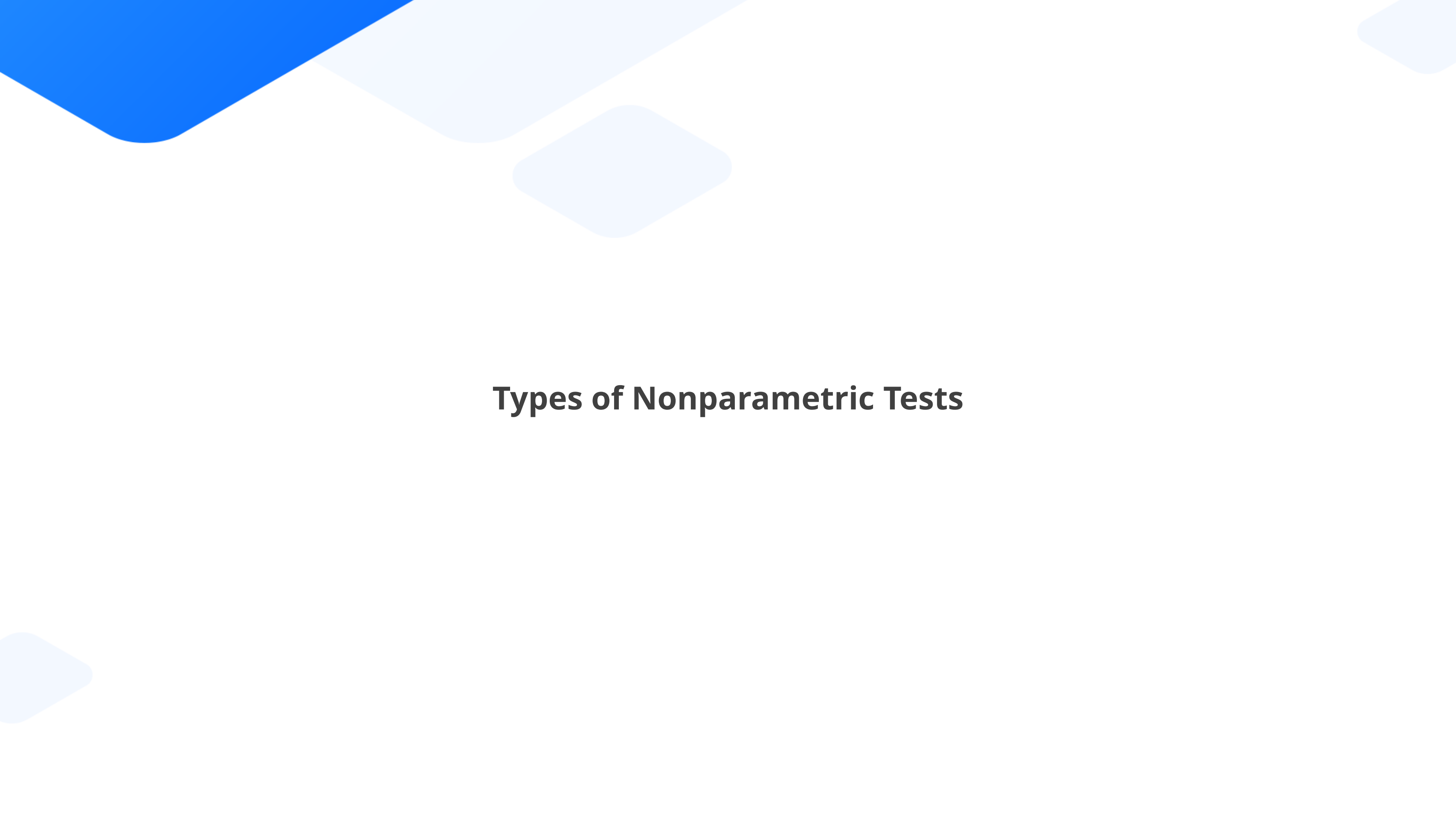
Thus, Alpha and Beta are kept low.

Discussion: Level of Significance

Duration: 15 minutes



- What is the level of significance?
Answer: The level of significance, also known as the significance level, refers to the probability of rejecting the null hypothesis when it is true. It is a crucial parameter in inferential statistics that assists researchers in deciding whether to reject or fail to reject the null hypothesis.
- What are the outcomes of hypothesis testing?
Answer: The following are the outcomes as below:
 - H_0 (null hypothesis) is true, and it is rejected.
 - H_0 (null hypothesis) is true, and it is accepted.
 - H_0 (null hypothesis) is false, and it is rejected.
 - H_0 (null hypothesis) is false, and it is accepted.



Types of Nonparametric Tests

Discussion: Chi-Square Test

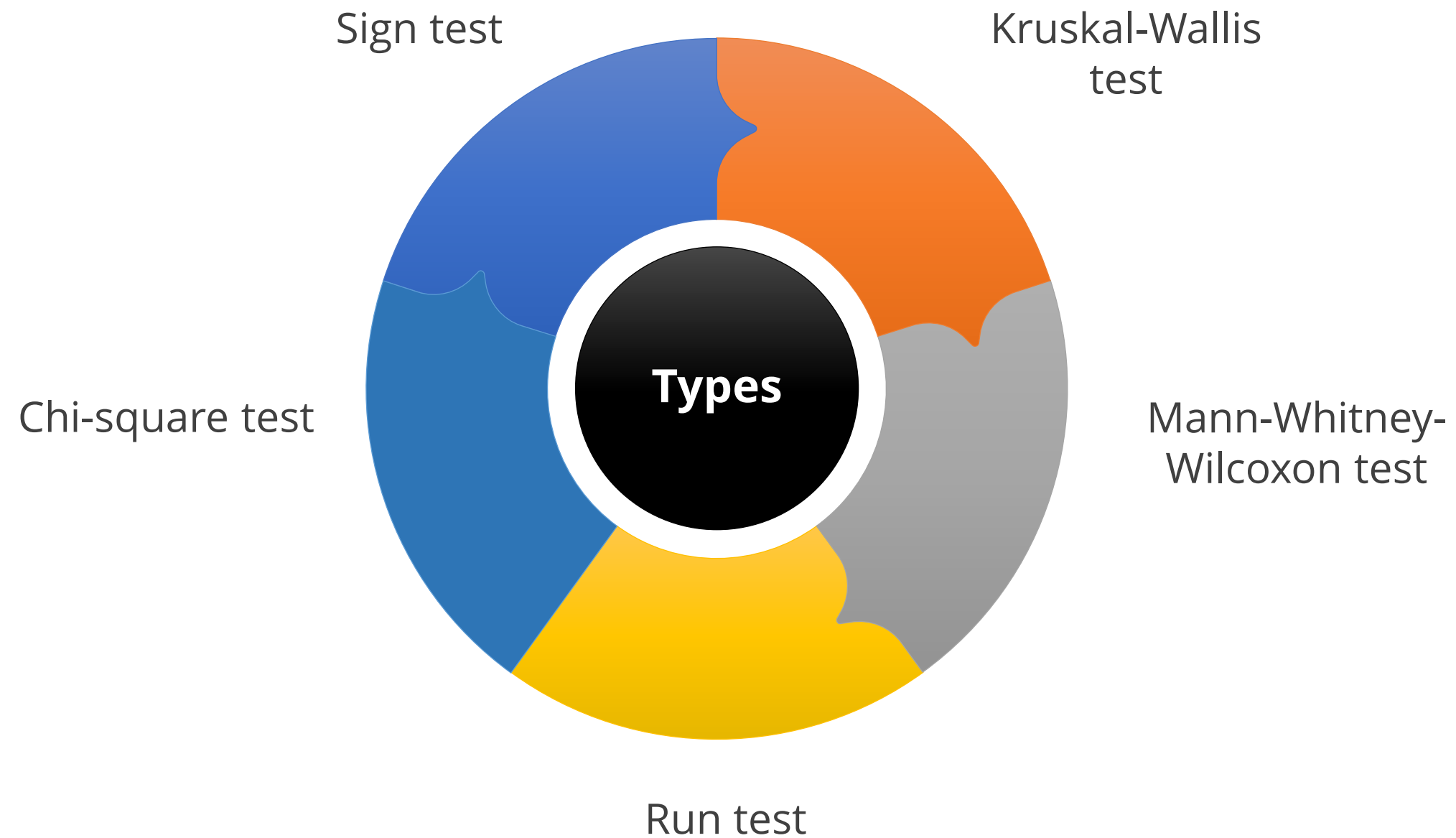
Duration: 15 minutes

- What is a chi-square test?
- What happens when two attributes are independent?



Types of Nonparametric Tests

Nonparametric tests do not require any assumption of a statistical distribution.



Chi-Square Test

A chi-square test is a statistical test used to compare observed and expected results.

The formula for chi-square test is given below:

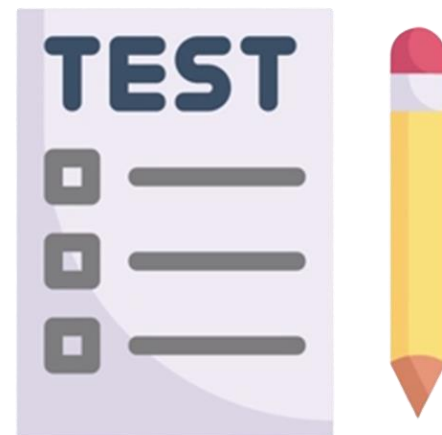
$$\chi^2_c = \frac{\sum (O_i - E_i)^2}{E_i}$$

- C= Degrees of freedom
- O = Observed Value
- E = Expected Value

It is used to test if the two attributes are independent or dependent.

Chi-Square Test: Example 1

The chi-square test is used to assess the significance of the relationship between mock test performance and subsequent improvement in the student's performance.



Mock tests



**Improve student performance
in board examination**

Chi-Square Test: Example 1

Mock tests and performance in the board examinations

Data (observed frequencies) collected from several schools

	Frequent	Seldom	Never	Total
Failed	12	18	45	75
Pass class	18	25	32	75
Second class	17	29	29	75
First class	22	24	29	75
Distinction	32	30	13	75
Total	101	126	148	375

Chi-Square Test: Example 1

Considering probability as a relative frequency:

The probability of
the student failing
 $= 75/375$

The probability of
the student
appearing for the
mock test =
 $101/375$

The value 75 represents the total number of failed students, while 101 represents the total number of students who frequently take mock tests.

Chi-Square Test: Example 1

Example: Mock tests and performance in the board examinations

	FREQUENT	SELDOM	NEVER	TOTAL
Failed	12	18	45	75
Pass class	18	25	32	75
Second class	17	29	29	75
First class	22	24	29	75
Distinction	32	30	13	75
TOTAL	101	126	148	375

$$\begin{aligned}\text{Expected frequency} &= \text{Row total} * \text{Column total} / \text{Grand total} \\ &= 75 * 101 / 375 \\ &= 20.2\end{aligned}$$

Chi-Square Tests

The following equation quantifies the overall deviation.

$$C = \sum (\text{Observed frequency} - \text{Expected frequency})^2 / \text{Expected frequency}$$

The summation extends across all cells.

When C exceeds a threshold value, reject the null hypothesis that the two attributes are independent.

Chi-Square Test: Example 2

The problem is to determine whether there is an association between smoking and lung cancer.

	Lung cancer	No lung cancer	Total
Smoker	150	170	320
Nonsmoker	60	250	310
Total	210	420	630

Chi-Square Test: Example 2

To calculate the expected frequencies for each cell in the table, use the following formula:

$$\text{Expected frequency} = (\text{Row total} * \text{Column total}) / \text{Sample size}$$

$$\begin{aligned}\text{Smoker with lung cancer} &= (320 * 210) / 630 \\ &= 106.66\end{aligned}$$

Chi-Square Test: Example 2

Use the following formula to calculate the chi-square:

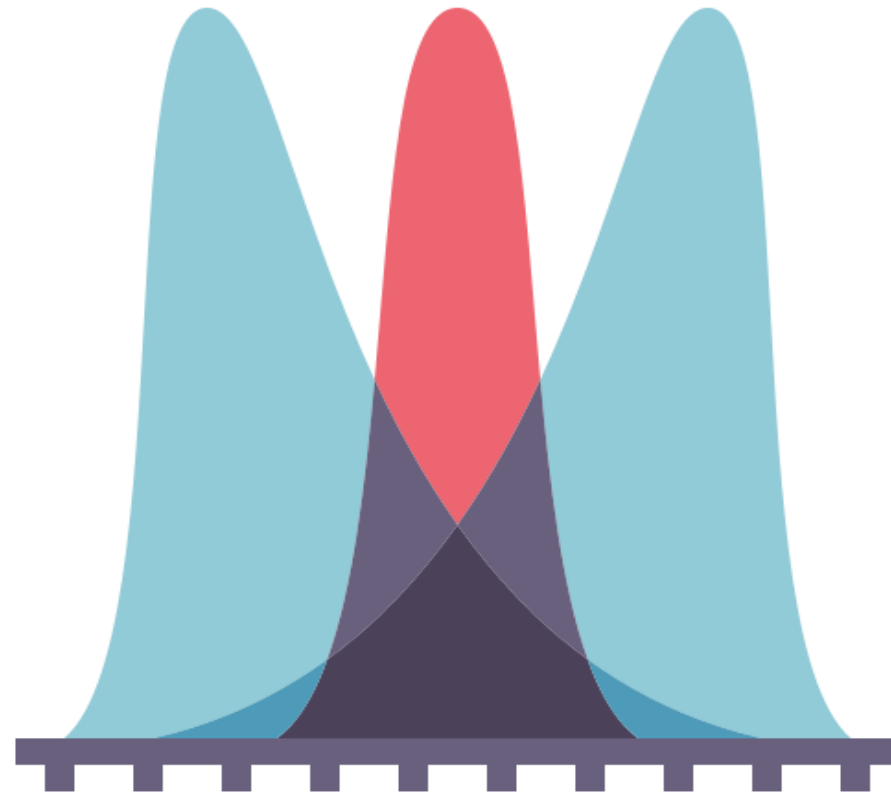
$$C = \sum (\text{Observed frequency} - \text{Expected frequency})^2 / \text{Expected frequency}$$

$$C = (150 - 106.66)^2 / 106.66 = 17.61$$

Note: Calculate the contribution from all other cells in the same manner, and then sum them up to obtain the final chi-square statistics

Fisher's Exact Test

The chi-square test is considered inaccurate when the expected frequency is less than 5.



Fisher's exact test is applied when the chi-square test is not appropriate.

Fisher's test, on the other hand, is based on a more advanced discrete distribution known as the hypergeometric distribution.

Discussion: Chi-Square Test

Duration: 15 minutes

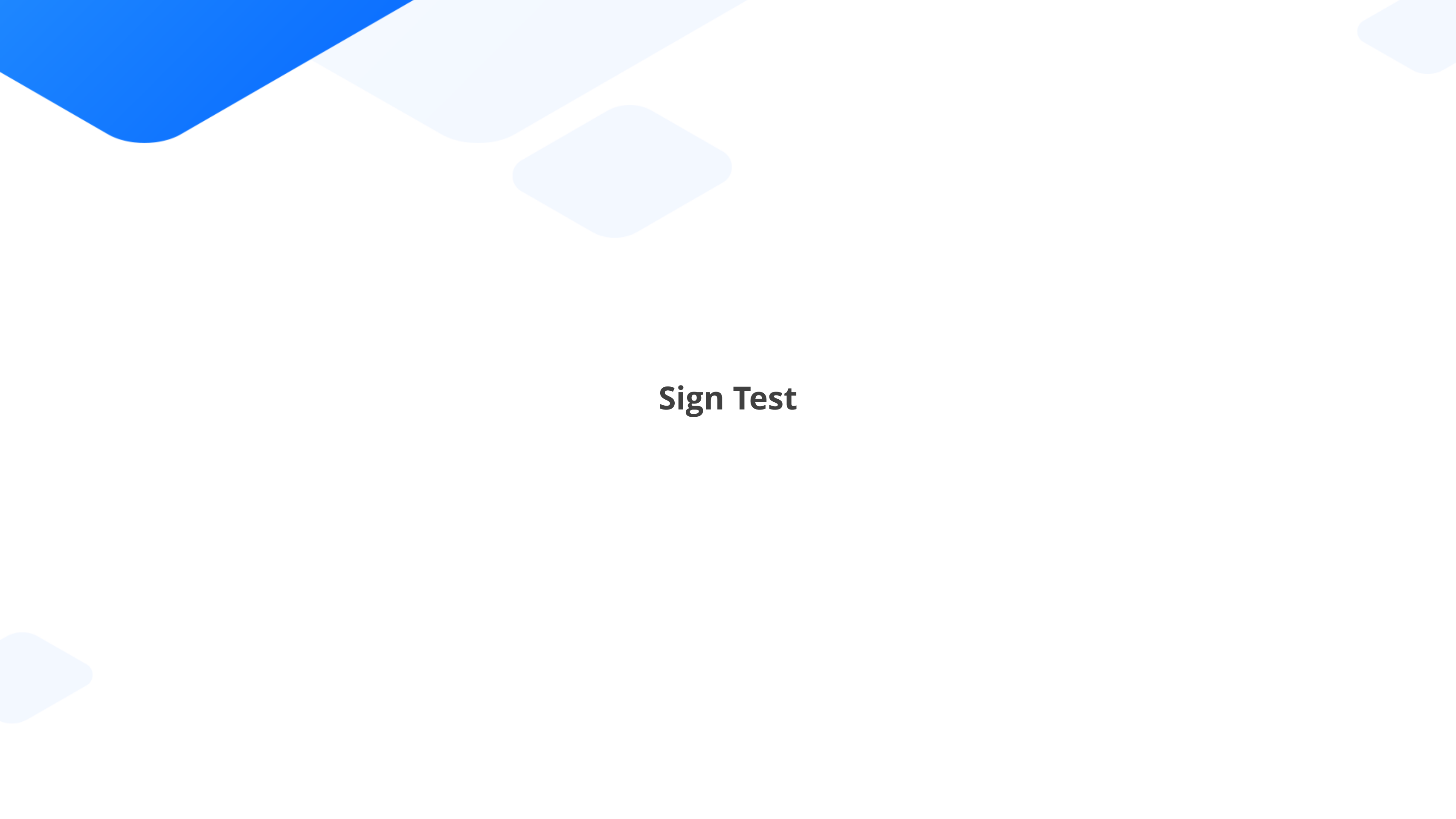


- What is a chi-square test?

Answer: A chi-square test is used to determine whether two attributes are independent or dependent.

- What happens when two attributes are independent?

Answer: If two attributes are statistically independent, there is no correlation between them. The value or occurrence of one does not impact or influence the other.



Sign Test

Sign Test

The sign test is utilized to evaluate the null hypothesis that the median of a distribution is equal to a specific value.

When dependent samples are paired and the bivariate random variables are independent of each other, the sign test is applied.

The sign test is based on the signs (plus or minus) of the observations, rather than their magnitude, and not vice versa.

It is often used as an alternative to the one-sample t-test when the data do not meet the assumptions of normality or when the sample size is small.

Sign Test: Example

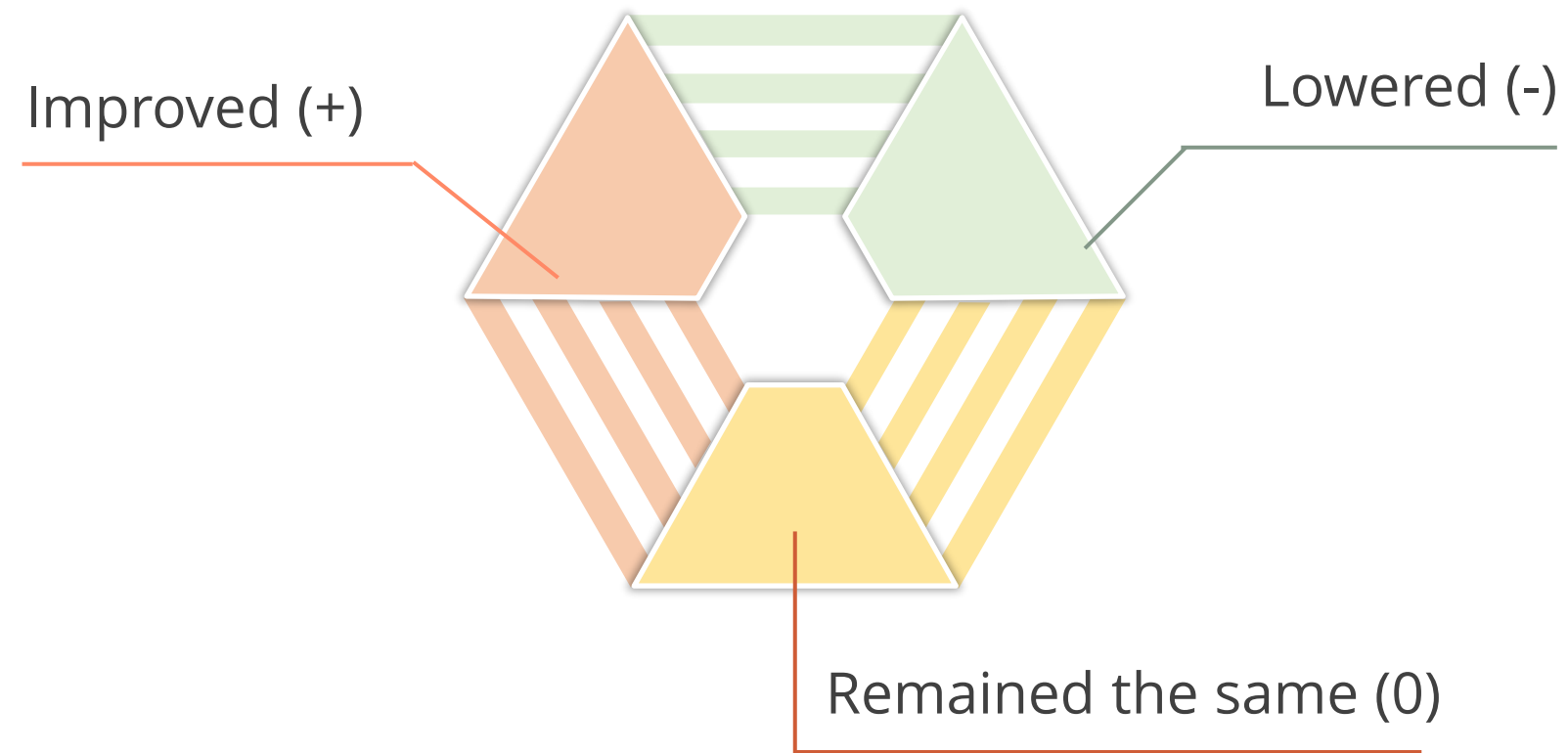
A mathematics tutor suggested that coaching could improve a student's performance on tests due to their teaching skills.



A group of 15 students was assigned to the tutor for two months of coaching, and their test scores before and after coaching were recorded.

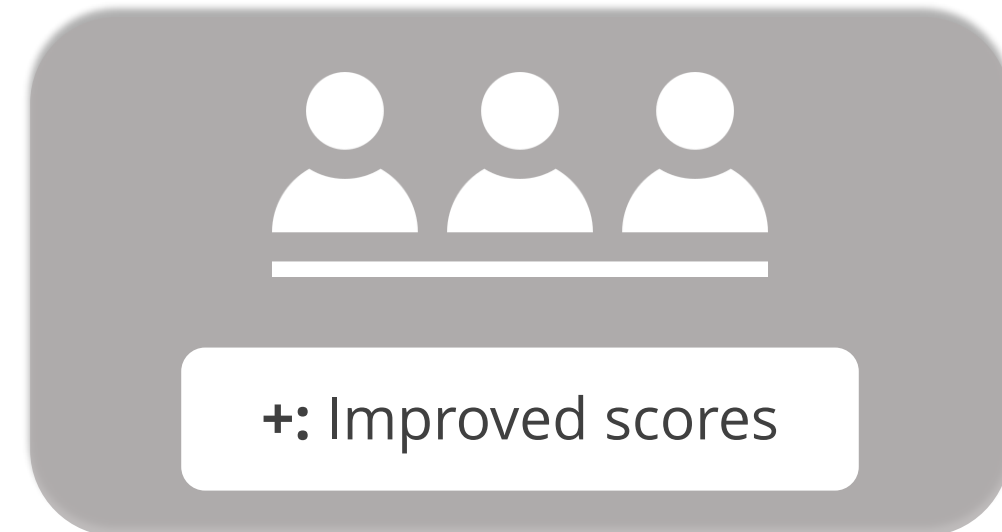
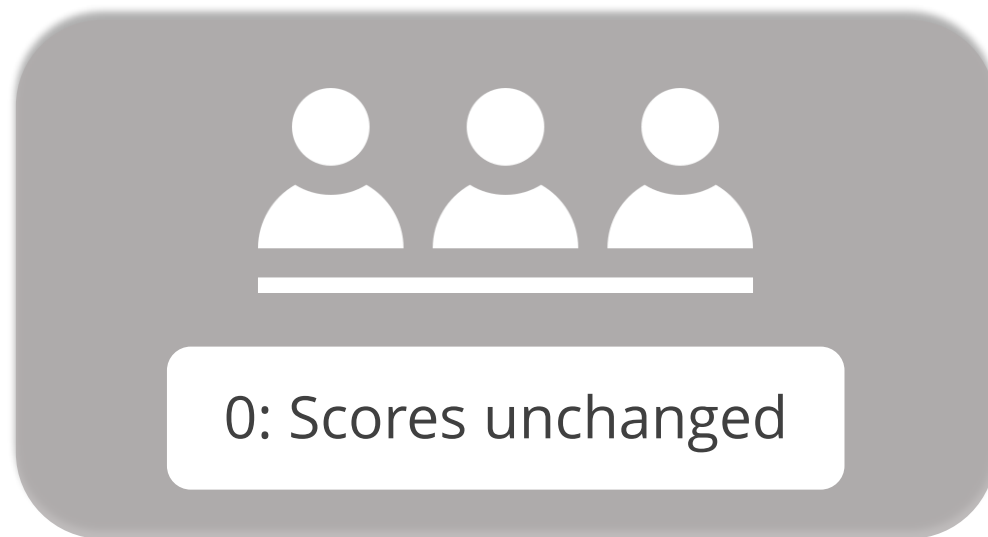
Sign Test: Example

For each student, +, -, and 0 are allotted according to whether score has:



Sign Test: Example

Here, students with scores unchanged are excluded.



The proportion of improved scores is determined.

Sign Test: Example

Null hypothesis

Coaching had no impact.

Proportion = 0.5

Alternative hypothesis

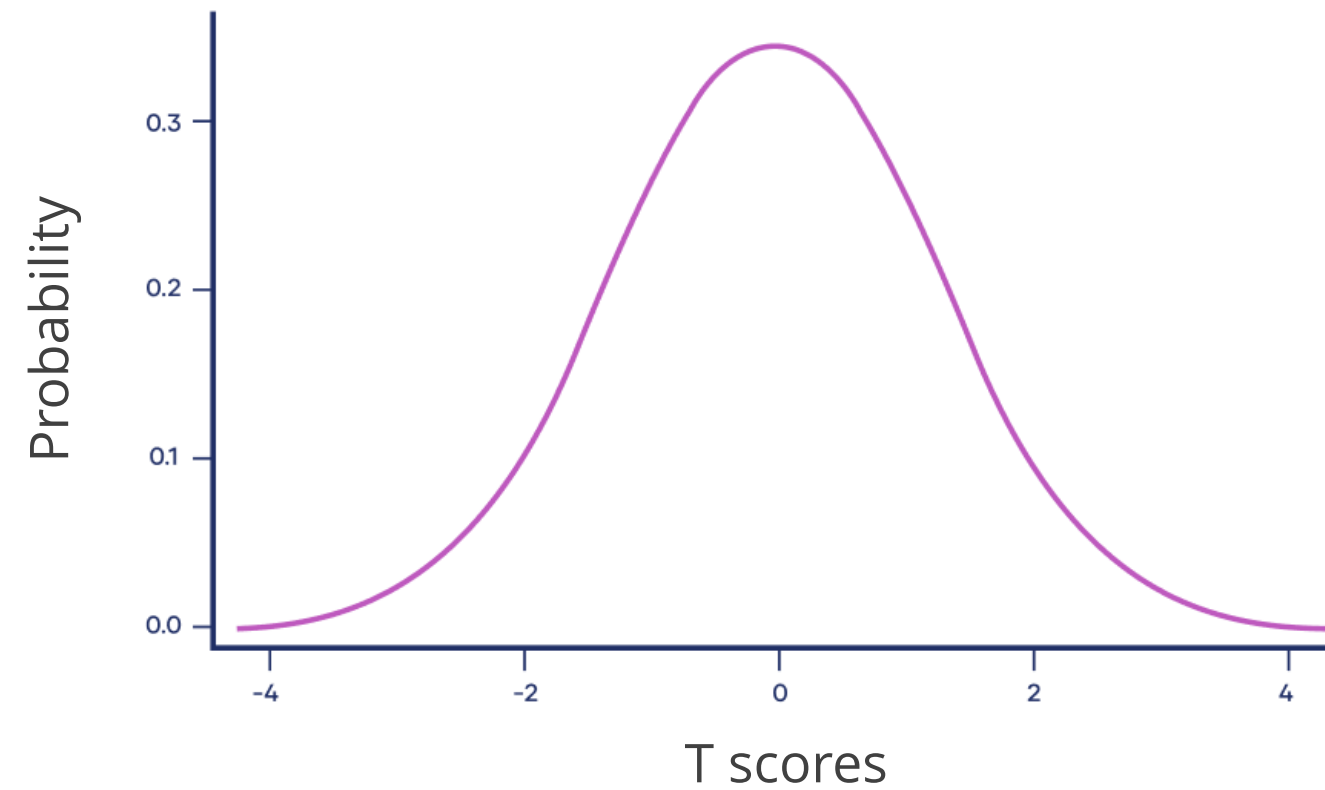
Coaching had a significant impact.

Proportion exceeds 0.5

When the sample proportion significantly exceeds 0.5, the null hypothesis is rejected.

T-Test

When dealing with a large sample size, the paired t-test can be utilized.

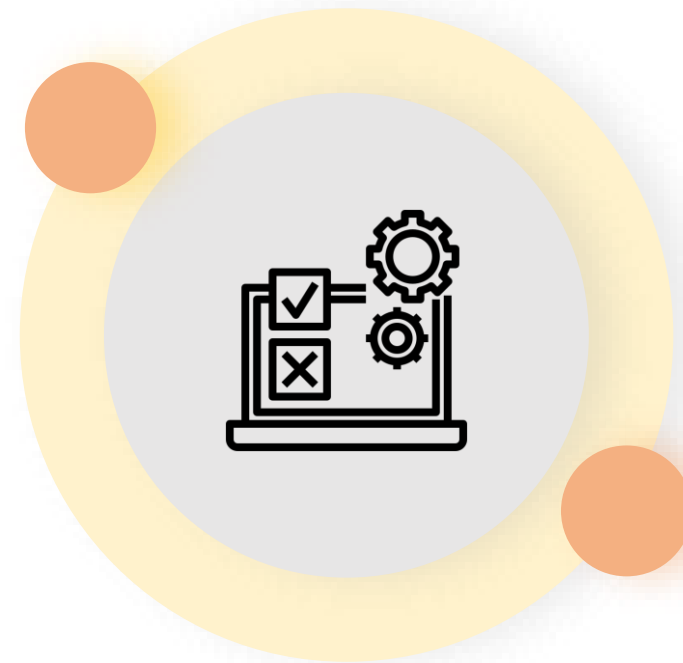


The t-test considers the magnitude of the difference in the student's scores to determine if there has been a change.

T-Test

Nonparametric tests are less sharp.

The paired t-test is more accurate.



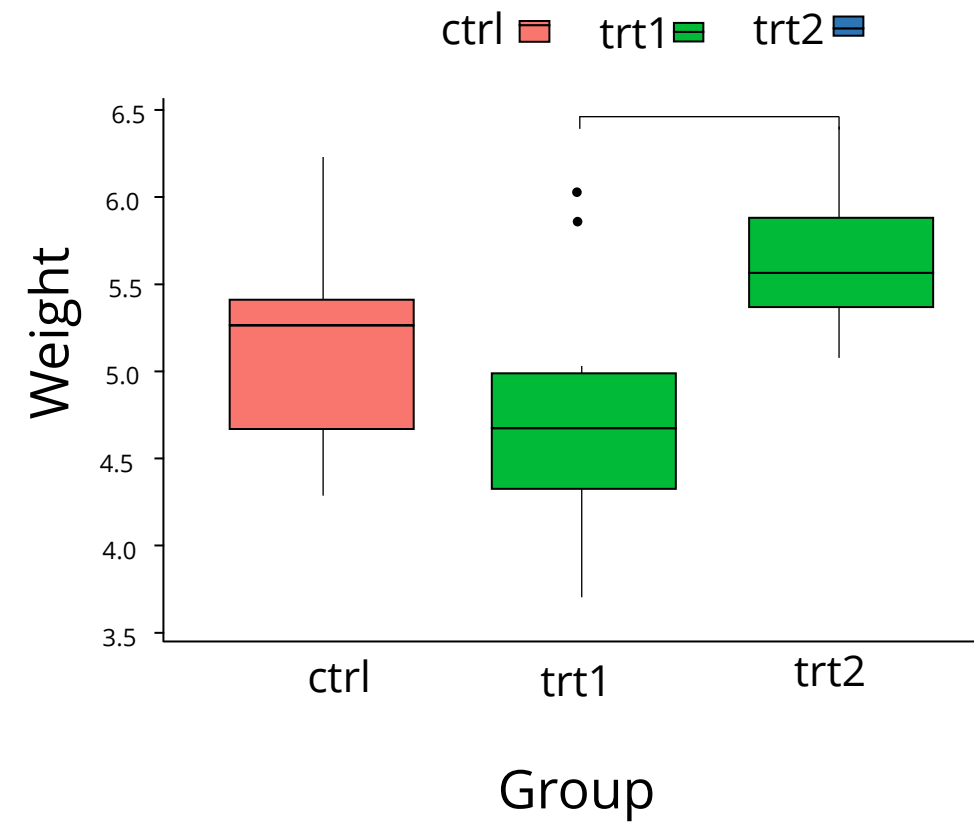
The paired t-tests require large sample sizes.

In assessing the effectiveness of coaching, it is sufficient to gather information on students' perceptions of improvement rather than focusing on specific score details.

Kruskal-Wallis Test

Kruskal-Wallis Test

The Kruskal-Wallis test is used when one-way ANOVA assumptions are not met, such as when the data are not normally distributed or the variances are not equal.



Kruskal-Wallis Test: Example

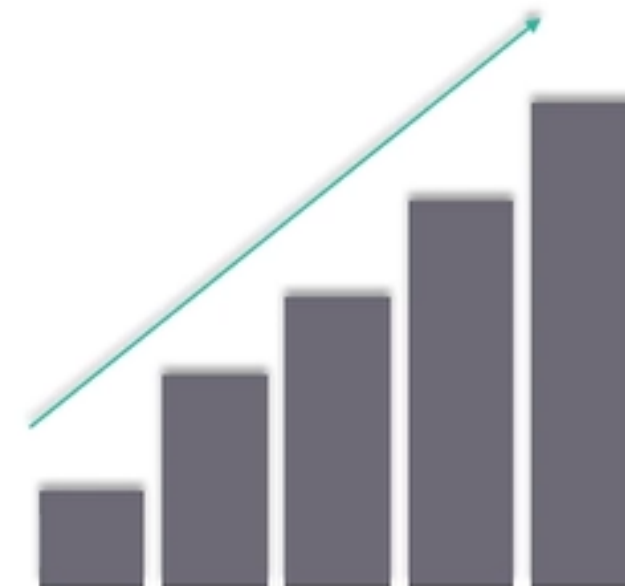
The number of passengers boarding buses on three different routes at 9 a.m. was observed for several days.



It is required to test whether mean numbers differ significantly.

Kruskal-Wallis Test: Example

Similar to the previous case, all the data is combined and the values are ranked in ascending order.



Ranks assigned

Kruskal-Wallis Test: Example

The total values of ranks are calculated for each of the three samples.



These values are considered insignificant when the population does not differ significantly.

Kruskal-Wallis Test: Example

To make an inference, the decision rule considers the values and sample sizes.

Unlike ANOVA, this test can be applied in the following scenarios:

Data does not need to be normal



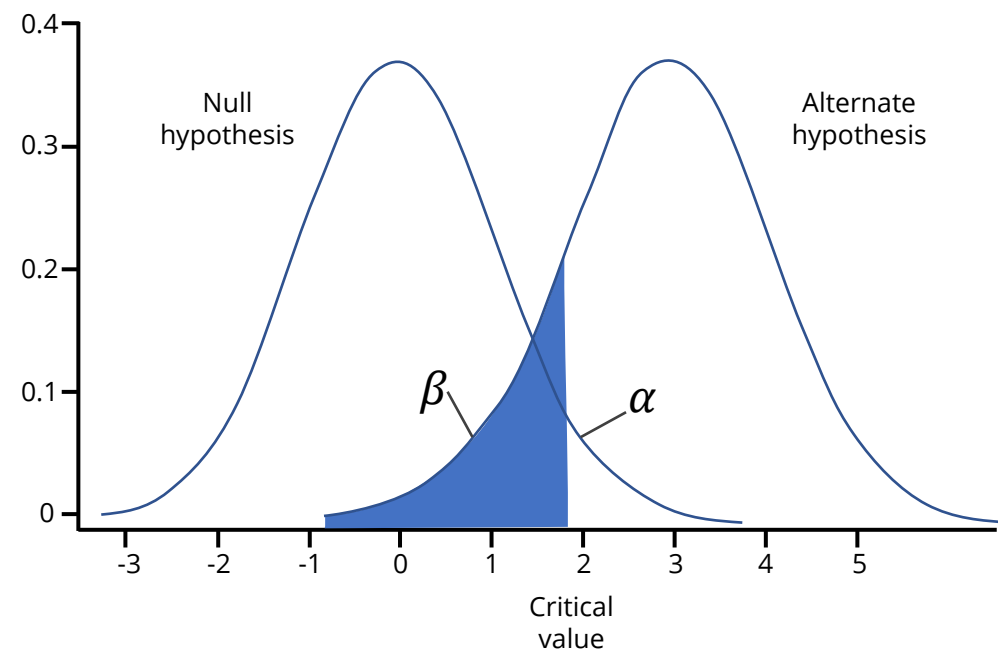
Sample size can be small



Steps for Kruskal-Wallis Test

Here are the steps involved in performing a Kruskal-Wallis test:

Step 1: State the null and alternative hypothesis



H0: The medians of all groups are equal.

H1: The medians of at least one group are different from the others.

Step 2: Rank all the observations, irrespective of their group membership, in ascending order from smallest to largest.

Steps for Kruskal-Wallis Test

Step 3: Calculate the sum of ranks for each group

Step 4: Calculate the Kruskal-Wallis test statistic with the following formula:

$$H = [(12 / N(N+1)) * \sum(R_i^2/n) - 3(N+1)]$$

Where:

N is the total sample size.

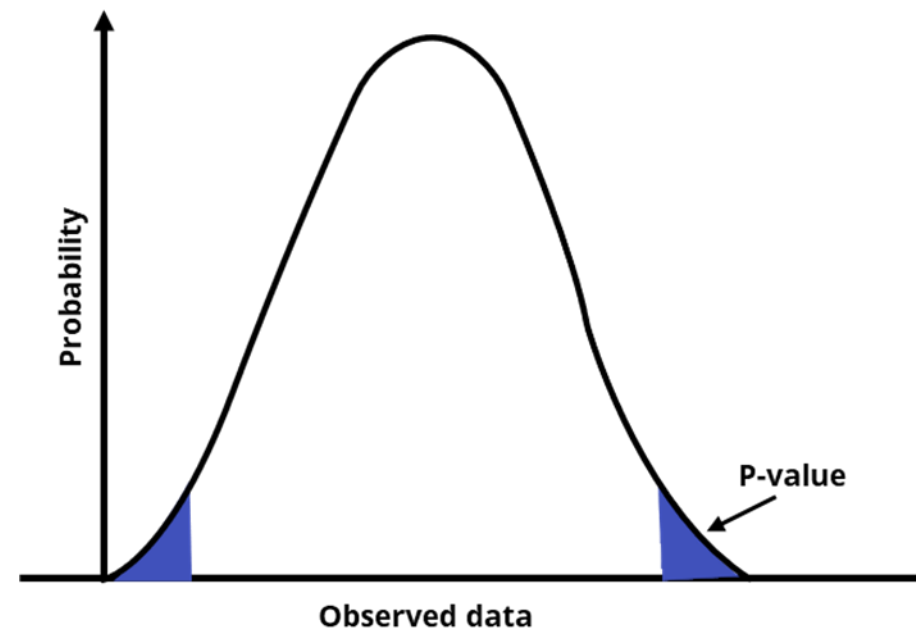
R_i is the sum of ranks for group i.

n is the sample size of group i.

Steps for Kruskal-Wallis Test

Step 5: Determine the degrees of freedom (df), which is equal to the number of groups minus one

Step 6: Calculate the p-value using a chi-square distribution with degrees of freedom



Step 7: Compare the p-value to the level of significance to determine if the null hypothesis should be rejected or not

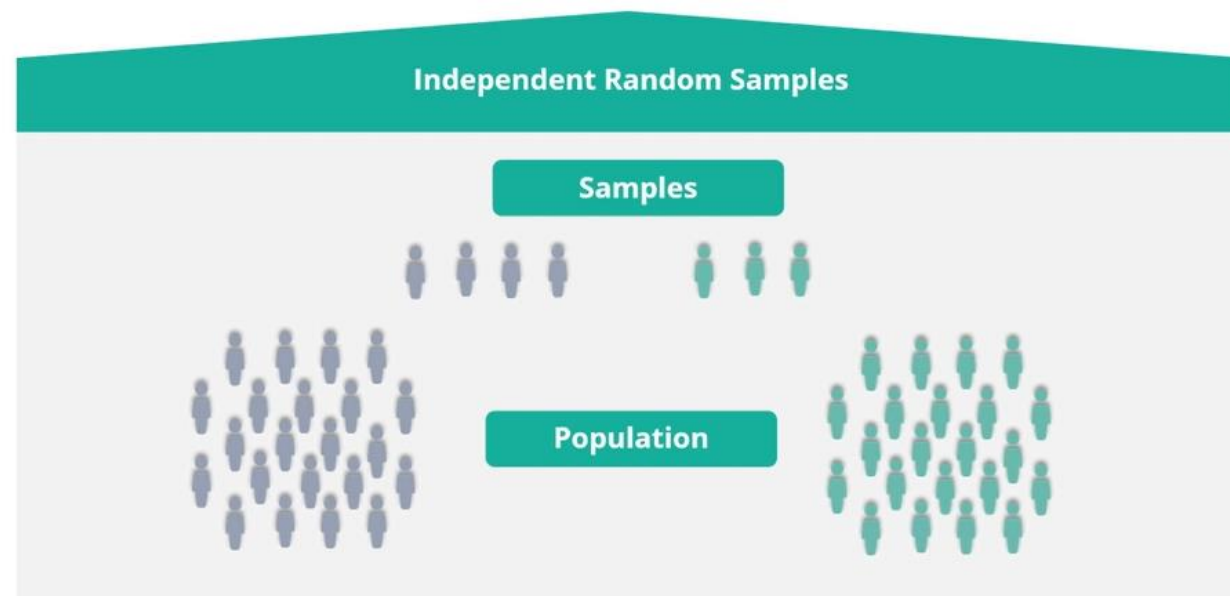
Note: If the null hypothesis is rejected, indicating a significant difference in medians among the groups, post-hoc tests can be performed to identify which specific groups exhibit significant differences from each other.



Mann-Whitney-Wilcoxon Test

Mann-Whitney-Wilcoxon Test

It is a nonparametric statistical test that is used to determine if there are differences between two independent groups in terms of their ranks on a particular variable.



Consider two independent random samples (possibly of different sizes) from two populations, as shown in the image:

The Mann-Whitney-Wilcoxon test can be employed after comparing the two populations.

Mann-Whitney-Wilcoxon Test: Example

Compare the ages of employees in two different divisions to examine whether ages differ significantly



Mann-Whitney-Wilcoxon Test

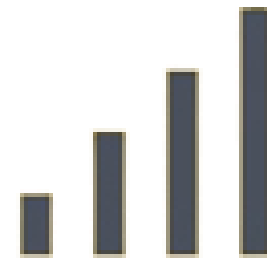
The ages of the random samples of both divisions are noted.



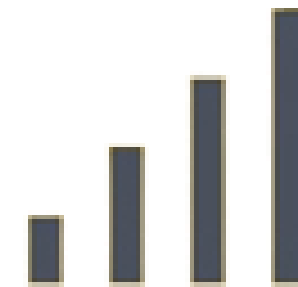
The two samples are combined, and observations are ranked in an ascending order of age.

Mann-Whitney-Wilcoxon Test

The sum of ranks for values in each of the two divisions is calculated separately.



Sum of rank values

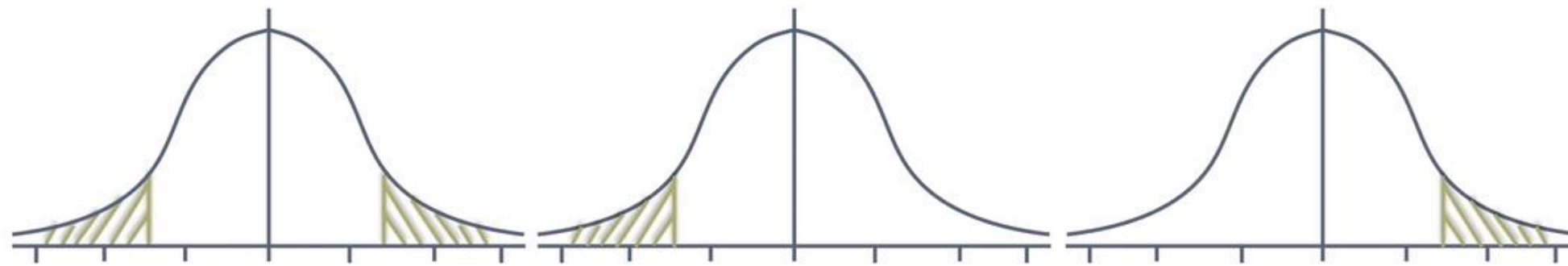


Sum of rank values

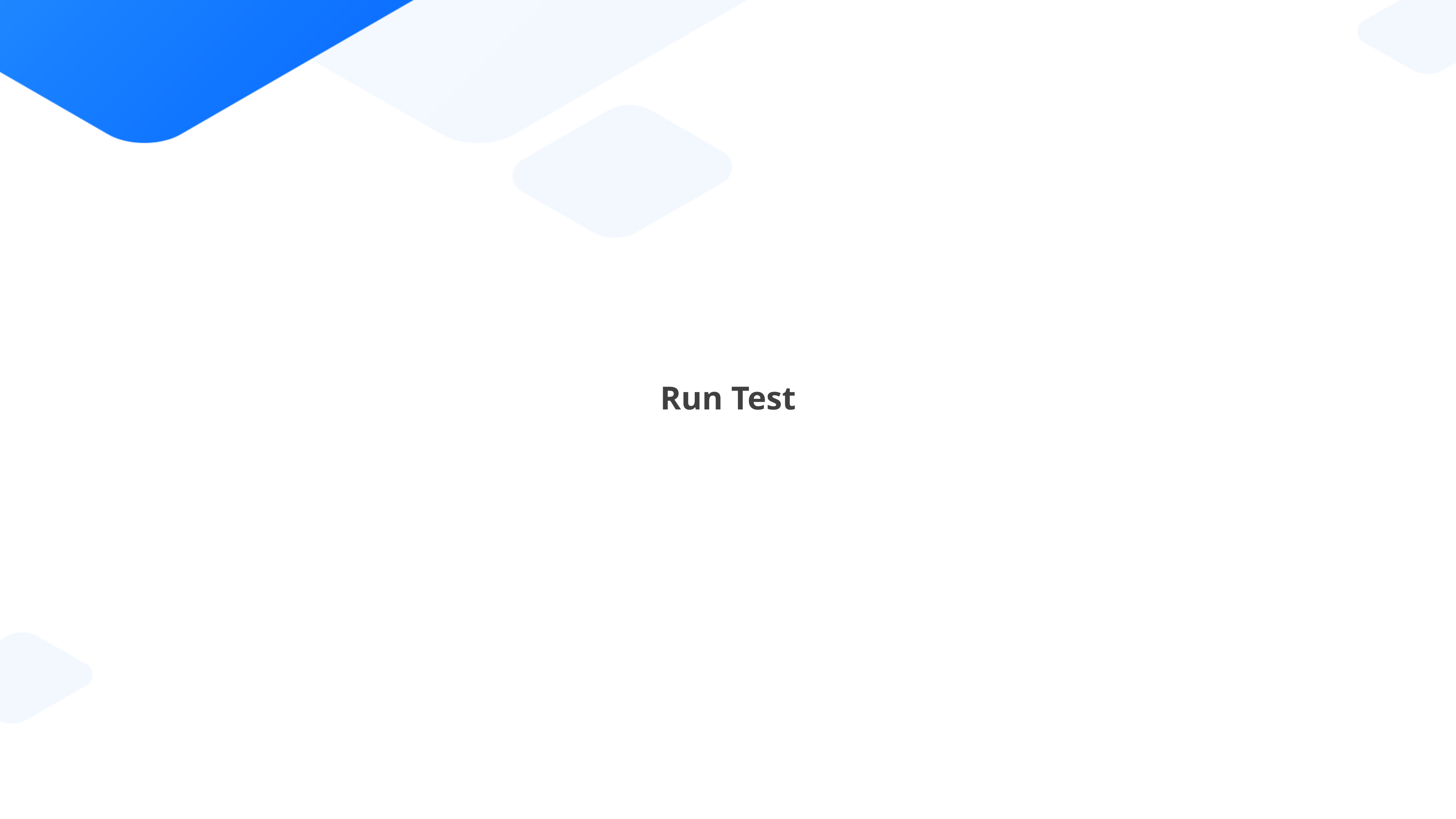
If these sums significantly differ, the null hypothesis stating that overall the ages do not differ significantly is rejected.

Mann-Whitney-Wilcoxon Test

The decision rules and threshold values depend on whether a two-tailed test, left-tailed test, or right-tailed test is used.



Similar to parametric tests, these choices are based on the alternative hypothesis.



Run Test

Run Test

The run test is a nonparametric test employed to ascertain whether there is an increasing or a decreasing trend in a time-ordered dataset.

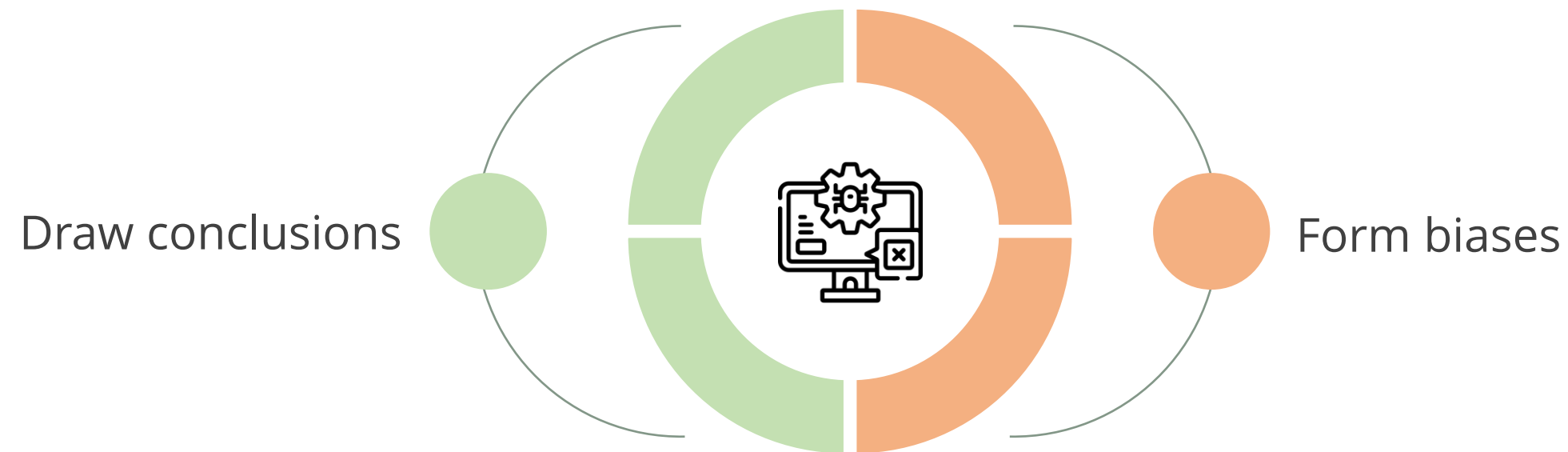
The two types of run tests are given below:

One-sided run test: This test is utilized to determine if there is a significant increasing or decreasing trend in the data.

Two-sided run test: This test is employed to determine if there is a significant trend in either direction, whether increasing or decreasing.

Run Test for Randomness

Data series sometimes tend to exhibit some patterns. The users of data tend to:



A run test is a logical, statistical test that can guard against incorrect conclusions or biases. The run test examines whether data occur in a random or non-random manner.

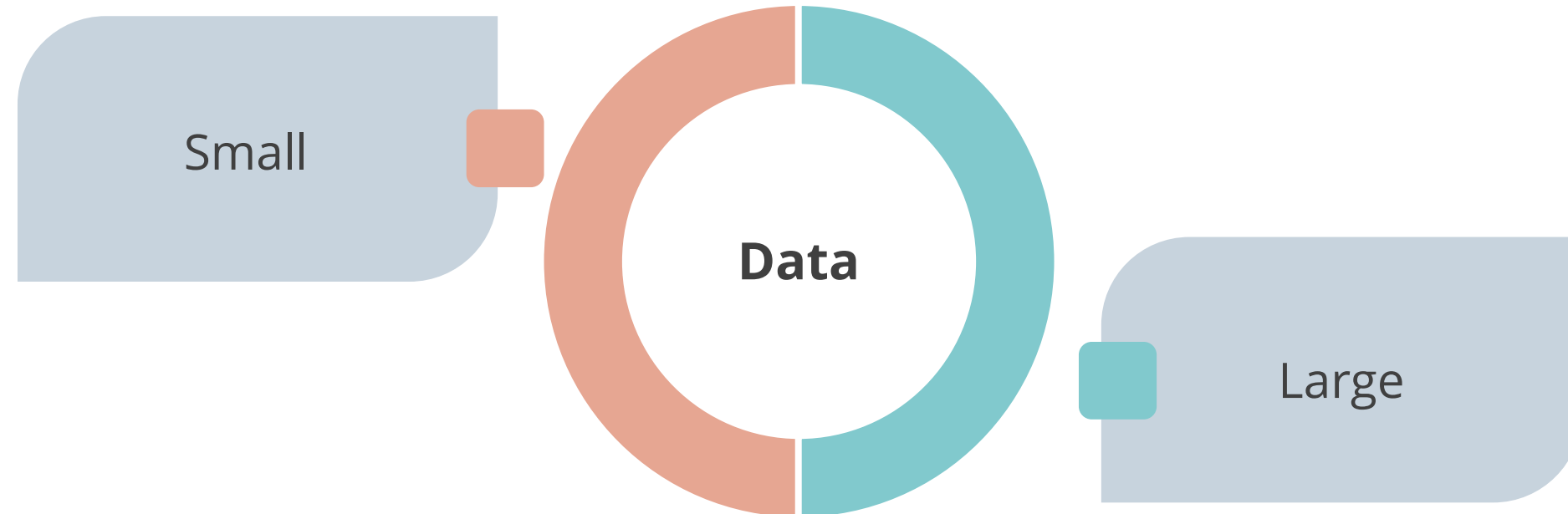
Run Test

In such patterns, a run is defined as a sequence of identical occurrences preceded or followed by other occurrences or none.



Run Test

The run test rejects the null hypothesis that states the data is random when the number of runs is either:



The threshold limit depends on the level of significance and the number of observations.

Run Test: Example

There is data collected on the number of customers visiting a store each hour for 12 hours. It is to be tested if the customer numbers are randomly distributed over time.

Here are the data:

6, 5, 3, 4, 7, 5, 4, 3, 6, 8, 4, 5

Step 1: Define the null hypothesis and alternative hypothesis

Null hypothesis: The number of customers is randomly distributed over time.

Alternative hypothesis: The number of customers is not randomly distributed over time.

Run Test: Example

Step 2: Calculate the total number of runs in the dataset

In this case, there are 6 runs in the dataset:

Run 1: 6

Run 2: 5, 3, 4

Run 3: 7

Run 4: 5, 4, 3

Run 5: 6, 8

Run 6: 4, 5

Run Test: Example

Step 4: Calculate the test statistic

$$Z = \frac{R - \left(\frac{2n_1n_2}{n_1 + n_2} + 1 \right)}{\sqrt{\frac{2n_1n_2(2n_1n_2 - n_1 - n_2)}{(n_1 + n_2)^2(n_1 + n_2 - 1)}}}$$

$$Z = (R - E(R)) / \sigma(R)$$

$$R = 6$$

$$E(R) = 6.45$$

$$\sigma(R) = 0.20122$$

The final value of $z = -2.236$

Run Test: Example

Step 3: Calculate the expected number of runs

$$E(R) = \frac{2n_1n_2}{n_1 + n_2} + 1$$

n_1 = The number of observations less than the median value

n_2 = The number of observations greater than the median value

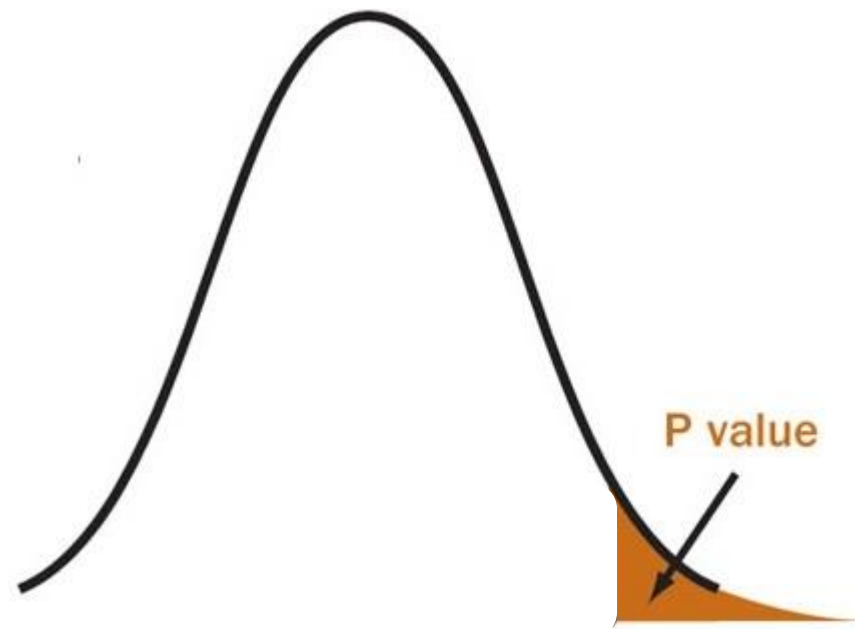
$$n_1 = 5$$

$$n_2 = 6$$

$$E(R) = (2 \cdot 5 \cdot 6) / (5 + 6) + 1 = 6.45$$

Run Test: Example

Step 5: The p-value for the test statistic of -2.236 is .012676, which is less than the significance level of 0.05.



Therefore, the null hypothesis is rejected, leading to the conclusion that the distribution of customers over time is not random.

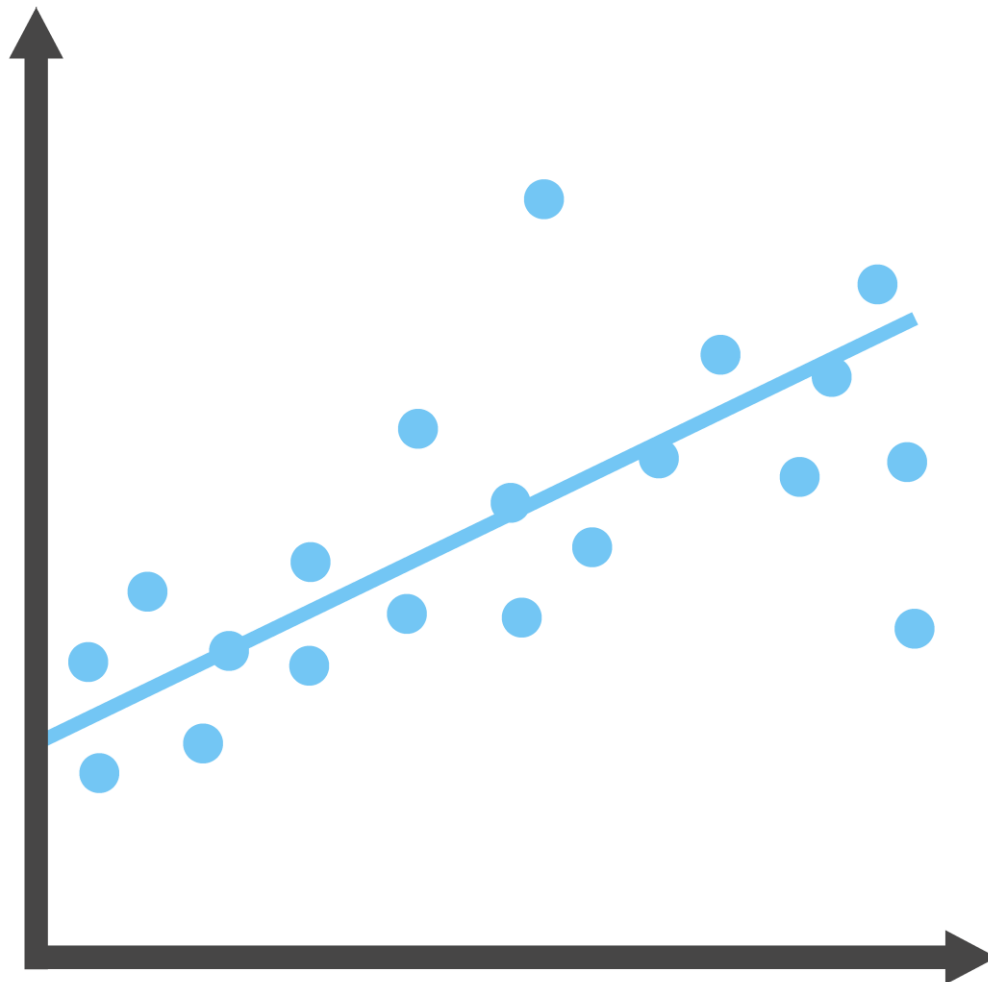
Note: Once the Z score is obtained, it is possible to find the associated p-value in a standard normal distribution table, or statistical software could be used for this purpose.



Regression Analysis

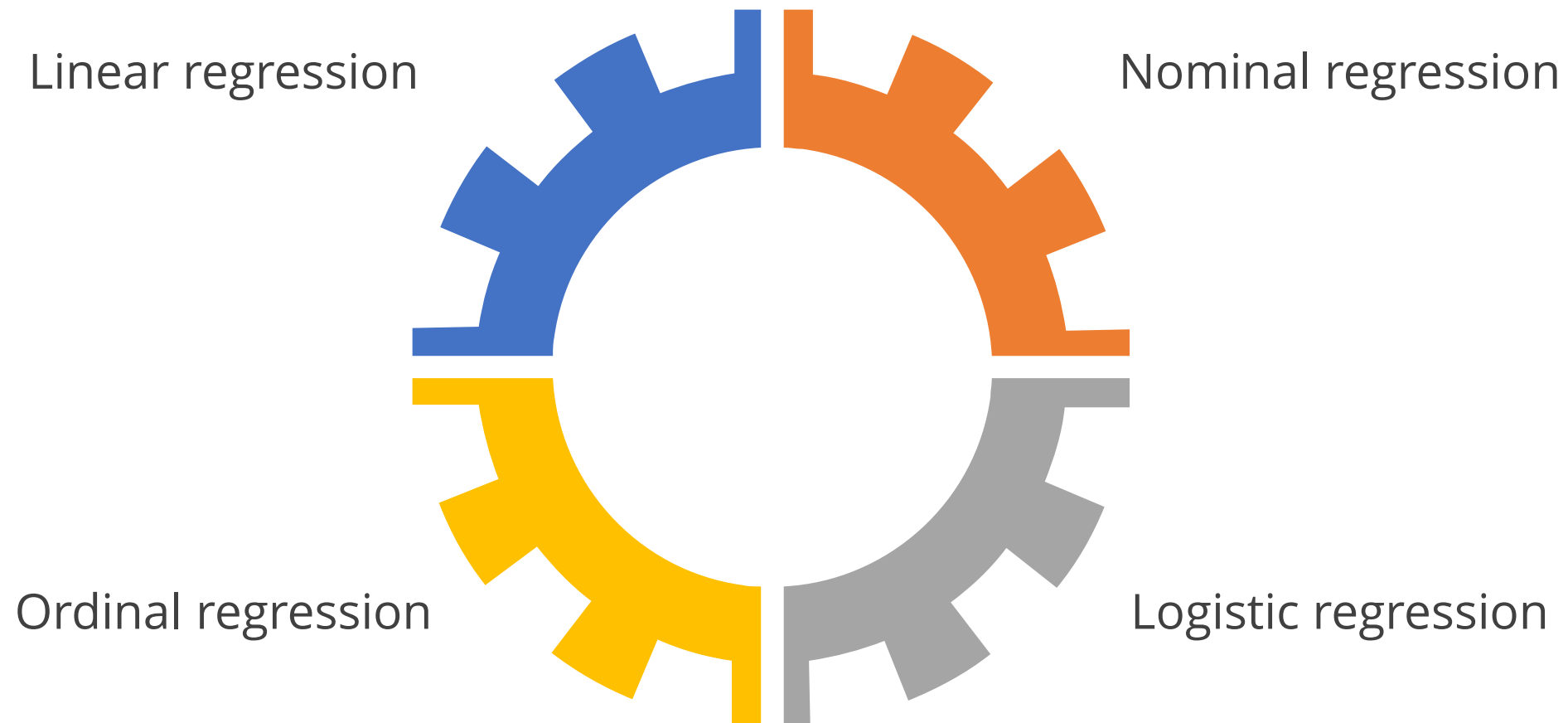
Regression Analysis

An analysis of regression assesses how one variable changes over time in relation to another.



Types of Regression Analysis

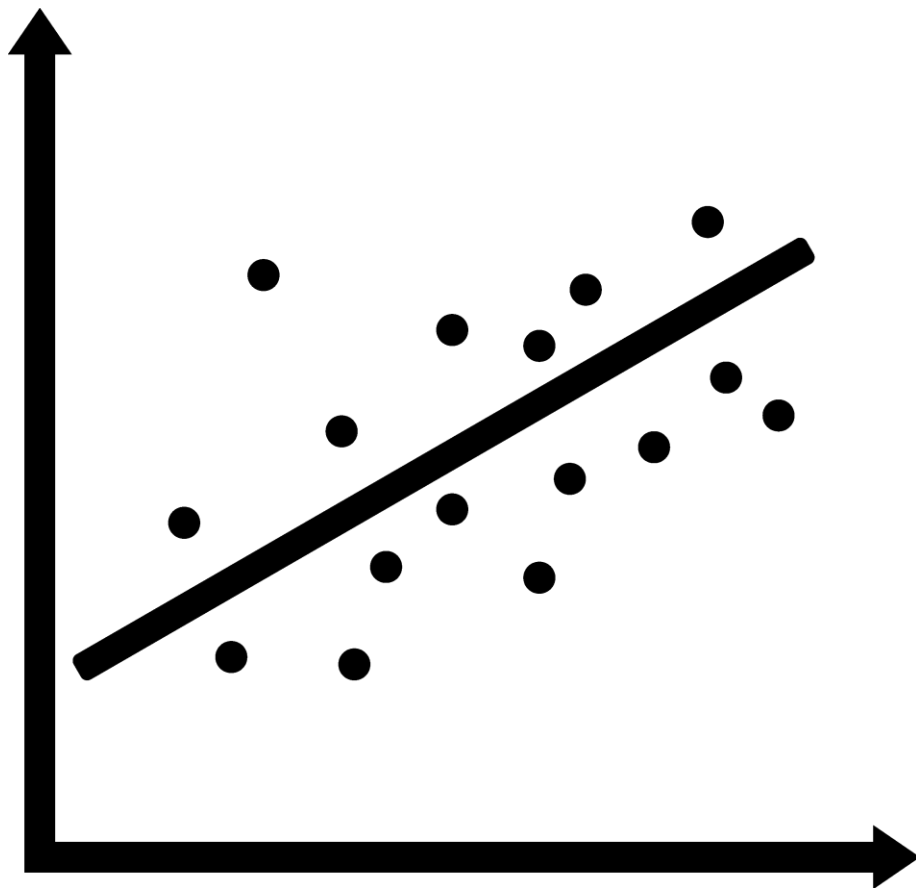
There are four types of regression analysis:



In inferential statistics, linear regression is the most employed type of regression.

Linear Regression

The linear regression method examines the effect of a unit change in the independent variable on the dependent variable.



The mathematical equation of linear regression is given below:

$$Y = a + bx$$

Where:

Y is the dependent variable

a is the intercept

x is the explanatory variable

b is the slope of the line

Linear Regression: Example

Collecting data on 100 employees, one can perform a linear regression analysis to investigate whether there exists a relationship between the number of years of experience and salary for the employees in a company.

Consider H_0 and H_1 as follows:

H_0 : There is no linear relationship between work experience and salary.

H_1 : There is a linear relationship between work experience and salary.

Level of significance: $\alpha = 0.05$

Correlation coefficient: $r = 0.85$

Hypothesis test: $t\text{-value} = 7.42$, $df = 98$, $p\text{-value} < 0.05$

Linear Regression: Example

Interpretation: The results indicate a significant linear relationship between work experience and salary.



The regression equation is $Y = a + bx$

Where:

Y is salary

x is work experience

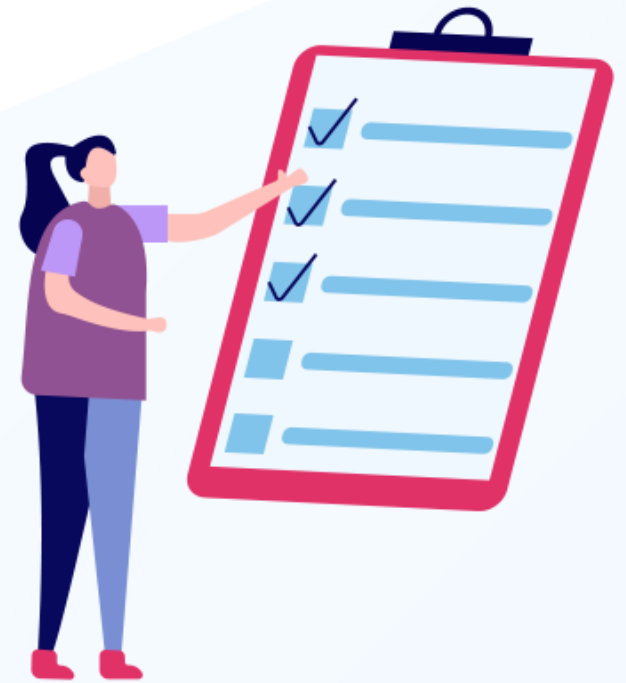
a is the y-intercept

b is the slope

The equation derived from the regression analysis can be utilized to predict an employee's salary based on their years of work experience.

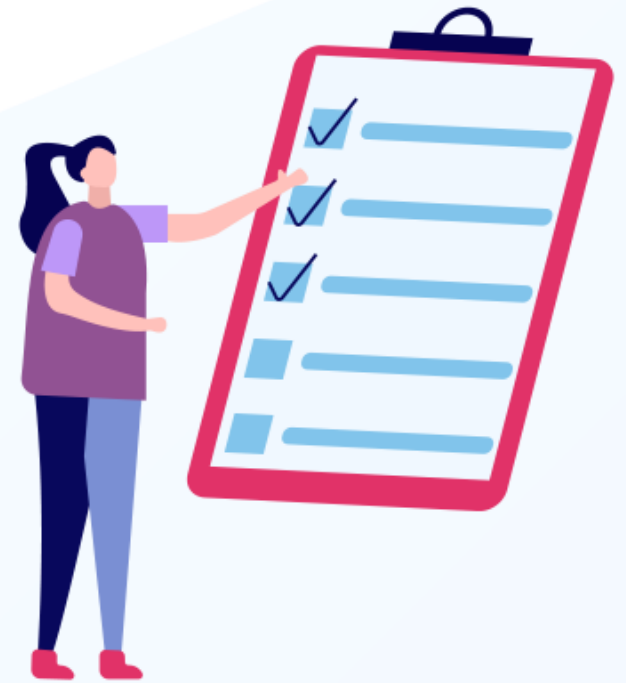
Key Takeaways

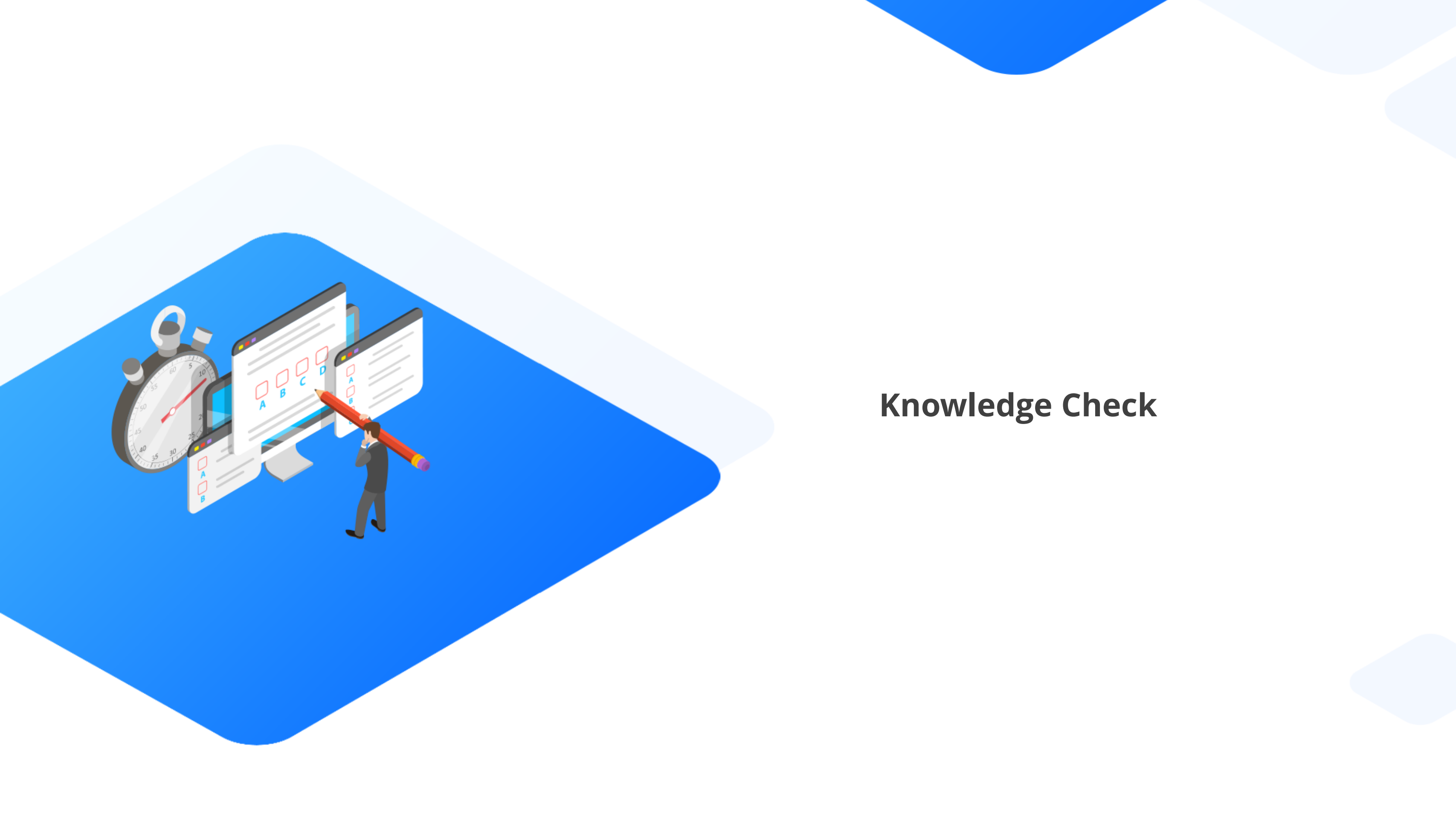
- Inferential statistics is a key tool in a data scientist's toolbox.
- In businesses, certain decisions are based on inferences from data.
- Statistical tests of hypothesis can be categorized into parametric tests and nonparametric tests.
- Various types of parametric tests are available, including the z-test, analysis of variance, Mann-Whitney-Wilcoxon test, Kruskal-Wallis test, and run the test for randomness.



Key Takeaways

- 👁️ A two-tailed test is applied to a normal distribution.
- 👁️ A one-tailed test is a statistical test in which the critical area of the distribution is one-sided, either on the left or right.
- 👁️ The p-value is an indicator of the ease with which one can reject the null hypothesis.





Knowledge Check

Knowledge Check

1

Which of the following is the negation of the assertions?

- A. Null hypothesis
- B. Alternative hypothesis
- C. Both a and b
- D. None of the above



Knowledge Check

1

Which of the following is the negation of the assertions?

- A. Null hypothesis
- B. Alternative hypothesis
- C. Both a and b
- D. None of the above

The correct answer is **A**

The null hypothesis is the negation of the assertions.



**Knowledge
Check**
2

Which of the following is an indicator of the ease with which one can reject the null hypothesis?

- A. P-value
- B. Type I error
- C. Type II error
- D. Z-test



Knowledge
Check
2

Which of the following is an indicator of the ease with which one can reject the null hypothesis?

- A. P-value
- B. Type I error
- C. Type II error
- D. Z-test

The correct answer is **A**

The P-value is an indicator of the ease with which one can reject the null hypothesis.



**Knowledge
Check**

3

Which of the following can be used to compare two independent random samples of different sizes from two populations?

- A. Kruskal-Wallis test
- B. Mann-Whitney-Wilcoxon test
- C. P-test
- D. T-test



Knowledge
Check

3

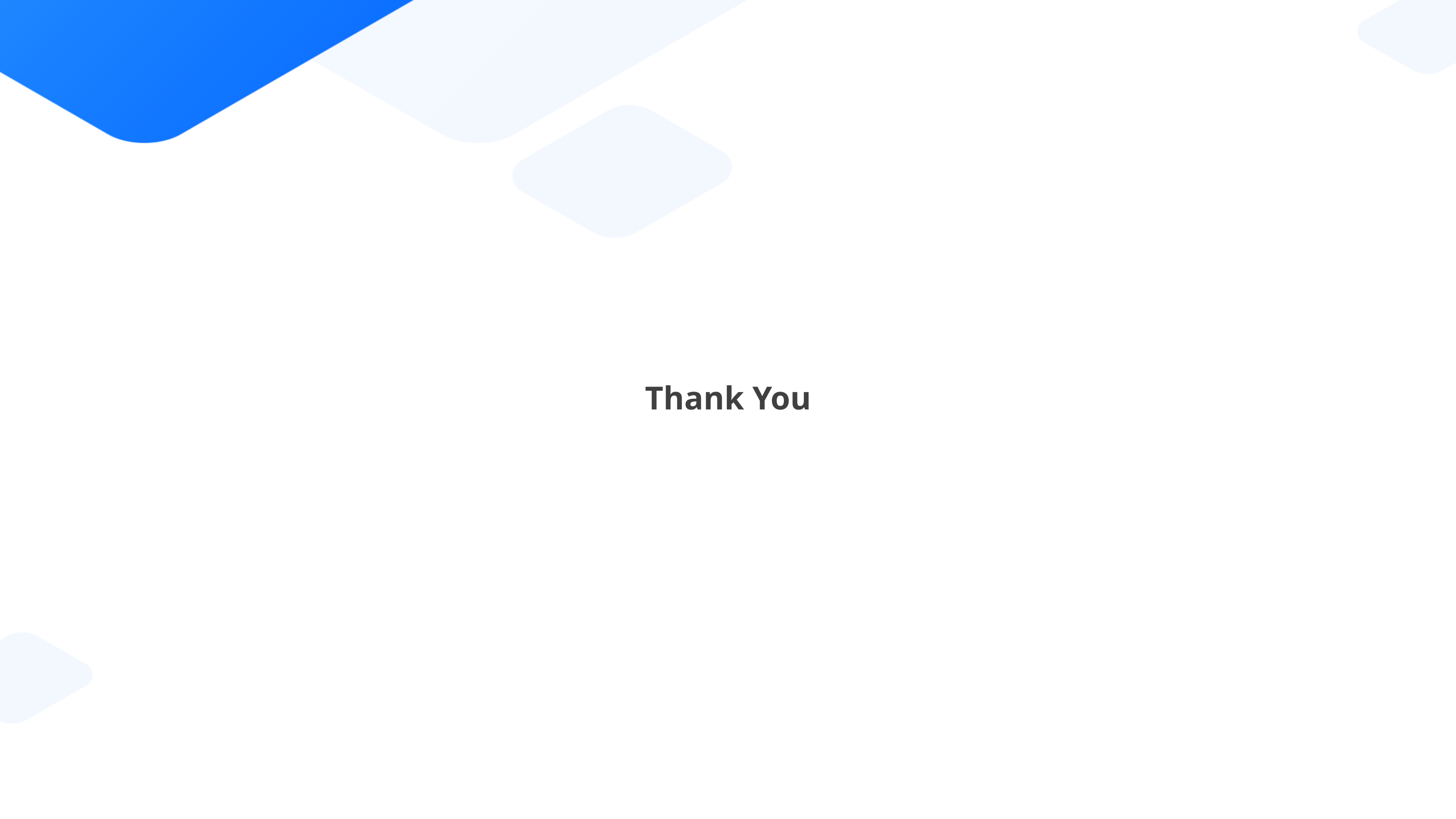
Which of the following can be used to compare two independent random samples of different sizes from two populations?

- A. Kruskal-Wallis test
- B. Mann-Whitney-Wilcoxon test
- C. P-test
- D. T-test

The correct answer is **B**

The Mann-Whitney-Wilcoxon test can be used to compare two independent random samples of different sizes from two populations.





Thank You